

THE DARK WEB AND HUMAN TRAFFICKING

By

Jamie Kay Graham

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ABSTRACT

This is a quantitative-comparative analysis that focuses on Artificial Intelligence (AI) platforms that assist law enforcement agencies as they combat human trafficking. Human trafficking is a Transnational Organized Crime (TOC) which means it can impact every country in the world, and in doing so, impact every person in the world. AI uses machine-learning capabilities to identify clusters, odd and/or unusual font, words, numbers, and other markers in advertisements that promote the sale of human beings. Human trafficking affects males, females, and children of all ages and can include different types of trafficking such as sex and labor trafficking. By using these AI platforms, law enforcement officers are able to identify and help more human beings than ever before in a quicker timeframe. This quantitative-comparative analysis compared Spotlight, Traffick Jam, Traffick Cam, and Domain Insight Graph (DIG) to determine if these platforms were helping law enforcement. The study revolved around the questions of accuracy, consistency, and effectiveness with each platform and found that the majority of AI platforms led the way to promote better, more efficient platforms by the same companies that learned how changes could assist law enforcement more in the future. While each platform assisted in their own ways, there were deltas in each area that leads to the need for future research in the area of AI and how it can be used to help victims of human trafficking and convict human traffickers more in later years.

Keywords: Human Trafficking, Sex Trafficking, Labor Trafficking, Transnational Organized Crime (TOC), Dark Web, Dark Net, Spotlight, Traffick Cam, Traffick Jam, Domain Insight Graph (DIG), Quantitative- Comparative Analysis, Artificial Intelligence (AI), machine-learning

Dedication

Former President John F. Kennedy said, “The rights of every man are diminished when the rights of one man are threatened” (Goodreads, 2022). I would not be where I am today if I did not follow those words. This dissertation is dedicated to all the people who are affected by human trafficking: the victims, the families, law enforcement, charities, research organizations, and anyone who is indirectly affected as well- so in a sense, every human being. We cannot, as a global unit, permit human trafficking to occur without combatting it with every piece of vigilance, heart, and soul that we have to deter and stop it. Each human being has a reason to live a free, just, and beautiful life that will not be scarred because of another who feels the need or want to abuse our human rights. I pray that each person who is affected by human trafficking finds peace and the victims know that they are not forgotten and they never will be.

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List of Abbreviations

AI	Artificial Intelligence
ASUCRP	Association of State Uniform Crime Reporting Programs
CSAM	Child Sexual Abuse Material
DHS	Department of Homeland Security
DOJ	Department of Justice
FBI	Federal Bureau of Investigation
GDP	Gross Domestic Product
HTRS	Human Trafficking Reporting System
ICE	Immigration and Customs Enforcement
iOCTA	Internet Organized Crime Assessment
IOM	International Organization for Migration
PREDAS	Peoples Recovery Empowerment Development Assistance
NIBRS	National Incident-Based Reporting System
NHTRC	National Human Trafficking Resource Center
PCA	Principle Component Analysis
SRS	Summary Reporting System
TIP	Trafficking in Persons (Report)
TOC	Transnational Organized Crime
TOR	The Onion Router
TVPA	Trafficking Victims Protection Act
TVPRA	Trafficking Victims Protection Reauthorization
UCR	Uniform Crime Reporting Program

UNODC	United Nations Office on Drugs and Crime
USCIS	United States Citizen and Immigration Services
VPN	Virtual Private Network

CHAPTER ONE: INTRODUCTION

Overview

Human trafficking is modern-day slavery which affects millions of people each year. Human trafficking does not only occur in a sexual manner such as prostitution or pornography but can include slavery in restaurants, drug cartels, tourism, street gangs, or in the health industry. While criminal justice systems and procedures have changed drastically throughout history, there are still many gaps that need to be addressed in national and international spectrums. In 2019, a study found that almost 900 pictures were uploaded to Instagram and over 76,000 videos were uploaded to be viewed on *YouTube* which indicated the need for more monitoring of social media (Fildago, et al., 2019). Human Trafficking is recognized as a Transnational Organized Crime (TOC) and has evolved with technological advances such as cyberspace which allowed human traffickers to reach people around the world with ease (Reid & Fox, 2020).

Cyberspace has led to many positive discoveries and connected people, businesses, and governments around the world. However, it has also led to many negative issues in the world of crime. The researchers stated that cyberspace has led to a human trafficking revolution in the last decade where it is not being conducted in a digital manner. Cyberspace globalization has increased anonymity and connections which makes it extremely difficult for criminal justice organizations to detect and track human traffickers (Reid & Fox, 2020). There is a need for additional studies to understand the new and evolving issues that come with the internet and human trafficking (Thomaz, 2020). These studies can describe the scope of illicit trades throughout the Dark Web (Thomaz, 2020).

The negative features of cyberspace has led to a more dangerous route which is referred to the Dark Web or Dark Net (Reid & Fox, 2020). Omar and Ibrahim (2020) stated that the Pentagon's secret network of connections that was developed in 1969 led to the establishment of the Dark Net. Unfortunately, the Dark Web is a fairly new development in cyberspace and it is used as a virtual black market for requests that are difficult to trace. The requests are difficult to trace because The Onion Router (TOR) is used to mask actions, criminal or legal, on the Dark Web (Omar & Ibrahim, 2020). Omar and Ibrahim (2020) discussed that normal internet platforms that are identified in the Clear Net do not connect to the Dark Web and the general public is usually unable to access the Dark Web. Because the lack of traceability in the Dark Net, human trafficking is able to increase with ease (Reid & Fox, 2020).

Background

Human trafficking has been advertised in many Clear Net platforms such as *Facebook* and *Craigslist* but classified platforms such as *Backpage* have been used to recruit and traffic individuals (Portnoff, 2018). Digital currency, Virtual Private Networks (VPNs), and TOR has led to an increase in trafficking of different products including human beings (Thomaz, 2020). With smartphones being used as personal computers, the safety of human beings is being threatened more as sales can happen at any time in any place that has access to the internet (Fildago, et al., 2019).

Criminal justice professionals have studied the linguistics of the internet, started to automate the classification of sex advertisements, and have started to categorize advertisements that have been authored by the same person or group of people, Artificial Intelligence (AI), and various algorithms but there must be more advancements (Portnoff, 2018). The context of this

study is based on the fact that cyberspace has led to the Dark Net which is a dangerous cyber platform for any individual that could be a victim of modern-day slavery (Portnoff, 2018).

Problem Statement

The Dark Net is a virtually untraceable internet platform and criminal justice professionals must be able to recognize human trafficking signs and be able trace the networks to help individuals who are being trafficked. Human trafficking has consistently changed with technological changes and while law enforcement professionals have been able to prevent it from happening in cases and protect and/or rescue individuals from the crime, there is always room for improvement (Reid & Fox, 2020).

The Dark Net should not be able to hide networks and connections from law enforcement authorities (Reid & Fox, 2020). Omar and Ibrahim (2020) stated that the Dark Web uses a high bandwidth and is a great distribution of objects to a limited amount of people who may copy anything if they believe it is interesting. Agencies must work together to develop strategies to share information and use tools to form data sets and determine patterns that exist or are developing as technology changes (Reid & Fox, 2020).

The Dark Net is a mysterious piece of cyberspace to most people in the world which is a major limitation for research (Omar & Ibrahim, 2020). Some people do not believe in the extent of dangers that the Dark Net presents while many people think it is more dangerous than what is known today. TOR and types of currency that are used on the Dark Net makes it highly improbable to trace most connections and networks currently (Omar & Ibrahim, 2020 & Fildago, et al., 2019).

Furthermore, innocent individuals are concerned about what will be traced with the algorithms which could infringe on their privacy by law enforcement or if their information can

be traced by other criminals for other crimes such as identify theft (Middleton, et al., 2017).

These fears are elevated when the knowledge of operational errors will most likely occur with algorithms and more sources are promoting and providing anonymity to anyone who wants to purchase those (Tai, et al., 2019 & Rabbany, et al., 2019). AI and cyber algorithms have been introduced into cyberspace, but information is limited, and the results are still being studied. The problem is that there is limited, to no, comparison of these platforms with each other which could potentially help identify shortfalls of others and determine what platforms are more successful in terms of accuracy, consistency, and effectiveness.

Purpose Statement

The purpose of this study is to compare different AI platforms that combat human trafficking. By doing so, the research should illustrate if programs are improving, lacking, or remaining constant in their purpose. Also, the study will shed light on if these specific programs are evolving with the crime.

The AI platforms that will be researched include *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*. They will serve as independent variables as the researcher will investigate the rates of success of apprehension and/or detection individually as well as dependent variables among each other as they are compared to each other in the study.

Significance of this Study

Human trafficking is modern-day slavery which means that it has been occurring throughout history. While it is not a new issue, the means that human trafficking is being conducted is constantly changing which indicates that this study is justifiable. Throughout this document, human trafficking will focus on the sexual trade because there is more research in the area, but it is important to remember that human trafficking can impact people through other

trades. The Clear Net has become a normal part of many lives, but the Dark Web is a questionable area that has gained momentum in the recent decades. This Dark Net promotes anonymity and evasion of criminal justice professionals which makes it extremely dangerous for the people, businesses, and governments that could be exploited (Reid & Fox, 2020). AI is being molded to combat the issues of human trafficking and provide law enforcement authorities a greater understanding of how the hidden parts of the internet operate and how to apprehend the criminals (Middleton, et al., 2017).

This proposed study will focus on how AI and algorithms have changed the criminal justice system in the pursuit of apprehending offenders and saving victims. The information in this study will be derived from public sources and the Dark Web will not be accessed to obtain any information.

Research Questions

RQ1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

RQ2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

RQ3: *In a span of six years, how effective detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

Definitions

1. *Artificial Intelligence (AI)*- Virtual technology that is used to trace patterns faster than a human (Wu, 2020)

2. *Algorithm* – A series of formulas in a virtual setting that is used to trace patterns (Wu, 2020)
3. *Black Market Forums* – Used to buy, sell, and trade items and, potentially, people on the Dark Web/Dark Net (Portnoff, 2018)
4. *Cyberspace* – The complete internet which would include the Clear Net and Dark Net (Reid & Fox, 2020)
5. *Dark Web/Dark Net* – A virtually untraceable area of cyberspace (Omar & Ibrahim, 2020)
6. *Mobile Applications (Apps)* – Virtual applications that can be used on desktop, laptop, smartphones, tablets, and other devices (Ortiz, 2019)
7. *The Onion Router (TOR)* – A router used in the Dark Web/Dark Net network to protect users from being tracked by their Internet Protocol (IP) addresses (Omar & Ibrahim, 2020).
8. *Transnational Organized Crime (TOC)* – A crime that transcends more than one country (Beasley, 2018)

Summary

This chapter discussed the background of human trafficking and the Dark Web and provided the justification on why this study is needed. Additionally, this chapter stated the problem, the context of the study, and the nature of the research. The research questions are presented in a simplistic manner yet will educate criminal justice professionals on the tools that are out there to assist them in their duties to protect individuals that could be victims of human trafficking. The assumptions, limitations, and technical terms are clearly defined to assist in the understanding that there are limitations and unfamiliar terms that many may not be familiar with.

The next chapter will be the literature review of previous sources and will discuss what is known and what is still unknown which led to the development of this study.

CHAPTER TWO: LITERATURE REVIEW

Overview

Human trafficking is a modern-day type of slavery where people are sold, exchanged, and exploited which benefits another individual or group in some type of monetary value (Konrad, et al., 2016). There is an estimated twenty-seven to thirty-five million people who are trafficked at any given time around the world which leads to approximately 150 billion United States Dollars (USD) profit for the traffickers (Konrad, et al., 2016). The Triad Mafia Groups of Asia and the Latin Cartels have been known to work together since 1993 where they traffic people into the United States of America (USA) at a rate that can range from 50,000 to 70,000 USD per person (Rajasegaran, 2017).

According to national data collected in 2017, human trafficking increased by 13 percent in one year, yet it is difficult to determine if the number of incidents overlapped or how many cases went unreported (Nsonwu, 2019). Rajasegaran (2017) explained that human trafficking is a crime that is committed by individuals, private organizations, and political governments around the world who use the black market which has spread to the Dark Net. This violation of the human rights is a global crisis and is increasing as advances in technology arise (Konrad, et al. 2016).

Technological advances such as mobile applications (apps) that are attractive to children have increased the TOC rates. For example, an app that children are using to chat with individuals has an icon that looks like a calculator to reduce the suspicions of parents, guardians, and other protectors. As the COVID-19 pandemic restricted physical movements of people, it encouraged virtual movements that allowed people to communicate online more than ever (Ortiz, 2021).

The UNODC stated that human trafficking is an issue everywhere in the world and the data that is collected may not be an accurate representation of the prevalence of the crime (United Nations, n.d.). The apprehension and conviction of offenders has begun to rise in recent years, but many areas are not showing the same amount of growth regardless of how much human trafficking is estimated to happen within them (United Nations, n.d.). This may be a product of the legislation in the individual governments and how well they are required to track criminal activity (United Nations, n.d.). Research and data are derived from cases that have been identified to better prevent new offenses and end existing crimes of human trafficking (Migration Data Portal, 2021).

Theoretical Framework

The Rational Choice Theory applies to the use of the Dark Web to commit criminal acts pertaining to human trafficking. Criminals use different means such as force or stealth to obtain their main goals, yet the Dark Web makes the traceability improbable. Regarding human trafficking, criminals do not feel empathy and show indifference to their victims (Beasley, 2018).

There are a variety of theories that researchers use to describe human trafficking issues. Templeton (2011) stated that a theory can relate to criminality through the measurement of crimes, offenders, and victims through different methodologies. The rise in substance abuse, sexual encounters for financial means, and child prostitution have made different theories apply to different parts of the world when discussing human trafficking (Lutya, 2017). This section describes how the General Strain Theory, Lifestyle Exposure Theory, and Routine Activities Theory could explain various parts of human trafficking.

The General Strain Theory

The General Strain Theory was introduced by Robert Agnew and can help describe the behavior of the victims, forced criminality, and additional knowledge pertaining to human trafficking. The General Strain Theory states that a person's immediate social surroundings is a critical factor and when Templeton (2011) paired it with human trafficking, they concluded that it was a determining factor concerning the likelihood that a person may become a victim of human trafficking. This theory can be measured through strain of emotions, behaviors, and cognitive processes that are altered as a result of human trafficking. The General Strain Theory is used to determine if sociological factors impact these individuals who will later engage in forced criminal acts. The General Strain Theory also impacts the family or those who live in impoverished areas who seek financial gain no matter the cost (Templeton, 2011).

Lifestyle Exposure Theory

The Lifestyle Exposure Theory was established in 1978 by Hindelang, Gottfredson, and Garofalo and is used to describe how a person's lifestyle could be responsible for their own victimization. This theory states that demographic characteristics such as race, age, sex, and a lack of adult guardianship could impact the likelihood that a person will become a victim to a crime. For example, if a female is targeted through a role expectation that she must be submissive to a male, she could be recruited into human trafficking (Lutya, 2017).

Routine Activities Theory

In 1979, the Routine Activities Theory was established by Cohen and Felson to describe how social interaction was a factor in the victimization of individuals. It is theorized the people are more prone to victimization when they are away from their home or when their routine activities are altered. If an offender seeks to obtain a person for human trafficking, they can study

the individual's behavior and abduct them at the most opportune moment. According to Lutya (2017), a person can become a victim through their work, leisure, sexual relationships, and/or social interactions. Furthermore, a person has less chance of escaping victimization because they are comfortable in their routine activities. The offender and the victim usually share similar lifestyles that do not have a consistent guardian monitoring their movements which improves the likelihood of the victimization (Lutya, 2017).

For victimization to occur, there must be a motivated offender, a lack of a capable guardian, and the target must be attractive to the offender (Lutya, 2017). Ingles (2019) stated that human trafficking brings in over nine billion USD annually which would make many offenders highly motivated where they can earn up to two-hundred thousand dollars per person. Regarding human trafficking, the motivated offender is the trafficker and the target is the trafficked individual (Lutya, 2017). Offenders will target a vulnerable person who lacks a guardian such as a police officer, parent, or friend in order to fulfill their goals through this theory (Lutya, 2017 & Ingles, 2019).

Related Literature

Human Trafficking

According to Beasley (2018), the United Nations Office on Drugs and Crime (UNODC) and the Protocol to Prevent, Suppress, and Punish Trafficking in Persons, also known as the Palermo Protocol, human trafficking are the actions involving recruitment, transfer, receipt of individuals, the transportation of the individuals, and/or harboring of these people through the means of force, abduction, deception, coercion, fraud, or abuse of power to exploit the vulnerability of an individual to obtain monetary benefits in the association with these actions. These actions may

include forced labor, sexual exploitation, organ removal for use on the black market, and other types of modern-day slavery (Beasley, 2018).

Many individuals become victims of human trafficking because the exploitation of economic or political hardships within their country which led to them fleeing the land only to be victims of exploitative criminals (Beasley, 2018). Rajasegaran (2017) referenced the Human Rights Watch Group who found that approximately one-hundred and fifty thousand immigrants were victims of the Japanese sex industry. Beasley (2018) explained that the victims may often find themselves willing to move drugs for Cartels and other groups that are involved in criminal activity because their hardships and vulnerabilities make them easy prey for false hope of a better life and then they are known as criminals for the drug trafficking which makes them apprehensive to contact law authorities afterwards.

It was not until the mid-1990s when the United States became more attentive to human trafficking when the Clinton Administration called for anti-trafficking efforts regarding females in European brothels (Horowitz, 2019). Former President William “Bill” Clinton addressed the “Three P’s” as the prevention of the crime, protection of the victims, and the prosecution of the criminals who trafficked the individuals (Horowitz, 2019). While the focus is usually on female trafficked victims, males make up approximately half of the victims of human trafficking in terms of laborious activities (Beasley, 2018).

Human trafficking is a Transnational Organized Crime (TOC) that threatens each person, government, and country through exploiting vulnerabilities of individuals and using the vulnerabilities to a criminal’s monetary advantage. These vulnerabilities could include immigration issues such as language deficiencies, economic hardships, natural disasters, or instability in the politics of their countries. According to Beasley (2018), forced labor and sexual

exploitation are the two main areas of human trafficking that is desired by criminals. Sexual exploitation usually occurs when people are deceived and later forced to perform sexual acts for the monetary gain of others. Individuals may be recruited through a false sense of friendly or romantic relationships, an inadvertent exposure to the sex industry, or through migration where they may be fleeing an unstable community. As immigrants migrate across borders, they become more vulnerable to being victims of human trafficking (Beasley, 2018).

Human trafficking is often associated with the trafficking of female individuals however males make up a significant portion of cases. When Samuel Vincent Jones, a faculty member and lawyer at the John Marshall Law School, studied a discussion on the *Larry King Live* media airing on CNN, he noted that masculine words were not used in the airing and females were the only sex mentioned. Unfortunately, this is a common trait as males are not usually viewed as victims of crimes pertaining to modern-day slavery. Contrary to this fact, males represent approximately half of the individuals who are trafficked for labor and have made up approximately 90 percent of child prostitutes in various counties (Beasley, 2018).

The International Organization for Migration's (IOMs) data analyst, Eliza Galos, performed a multi-level logistical regression and discovered that demographic profiles of males who are isolated or alone, are less educated, are extravagant spenders, have fled conflicts, or spend long times travelling are more likely to become victims of human trafficking (Beasley, 2018). According to Beasley (2018), the males who lack necessities but are physically strong and younger in age are targeted because of their potential value as an effective drug trafficker in dangerous terrains.

Unfortunately, victims of human trafficking are less likely to report the crimes because their lives or the lives of their relatives are threatened by death or serious injury. Furthermore,

victims may be subject to detainment, prosecution of crimes they were forced to commit while they were owned by another party or because they violated laws pertaining to immigration (Beasley, 2018).

Legislative Moves against Human Trafficking

The Palermo Protocol was established on November 15, 2000, by the United Nations and it was used to supplement the United Nations Convention against Transnational Organized Crime. While the United Nations adopted it in 2000, the United States would not truly implement the Palermo Protocol until November 3, 2005. Unfortunately, this is part of the issue with tracing human trafficking offenses as each nation state is able to determine what the protocol means and how it is used in their jurisdictions (Beasley, 2018).

The Trafficking Victims Protection Act (TVPA) would also be introduced to the United States in 2000 and addresses slavery, the sex trade, and involuntary servitude. While the TVPA was used to criminalize the act of human trafficking, it did very little to address domestic human trafficking in minor offenses and the victims were further victimized by the criminal justice system through dismissal of their needs or treated as criminals themselves. The Trafficking Victims Protection Reauthorizations Act (TVPRA) was introduced in 2005 to combat the derogatory treatment of victims of human trafficking (Beasley, 2018).

Many countries have improved their legislation to prevent human trafficking, prosecute the traffickers, and assist victims after they are rescued from their hostile environment (Beasley, 2018). Unfortunately, data still reflects the identification of victims more than how to prevent and trace criminal activity pertaining to human trafficking (Beasley, 2018). The most effective ways to combat human trafficking include descriptive statistics and the interaction of multiple criminal justice agencies on a local, national, and international level (Beasley, 2018). United

States Federal agencies such as the Federal Bureau of Investigation (FBI), Immigration and Customs Enforcement (ICE), and the Department of Homeland Security (DHS) have the most amount of data pertaining to crimes involving human trafficking (Farrell & Reichert, 2017).

In 2012, there were three hundred and six FBI investigations, yet this number is surprisingly low as many crimes are not reported to the federal level or are not reported at all. Federal agencies do not have jurisdiction unless the crime crosses state lines nor do they have the resources to support local criminal justice agencies in each case that arises. While the Federal Strategic Plan was developed to aid local jurisdictions, the local agencies understand their environment more and should be able to process the majority of information and ask for Federal assistance if the need presents itself. Local criminal justice agencies are the first responders to the crime and may have insider knowledge that the Federal agencies cannot obtain without the help of the local agencies (Farrell & Reichert, 2017).

The Federal Government of the United States has developed task forces that unite agencies around the nation in the processes of identifying and responding to crimes associated with human trafficking. Since 2008, there have been an increase in the details within human trafficking investigations with the establishment of the Human Trafficking Reporting System (HTRS). The HTRS identified over seven thousand people who could have been a victim of human trafficking between the years of 2008 and 2014. Unfortunately, the HTRS does not represent the nation and only a limited number of agencies were reporting the cases which means the number of cases could be much higher than the database suggests. Additionally, the database is only as good as the information that is presented. Many cases may not include complete details which skews the complete description of the cases (Farrell & Reichert, 2017).

Between 2010 and 2012, the state of California reported over twenty-five hundred cases involving human trafficking but less than thirteen-hundred identified victims. While some states employ local task force agencies, the information may be incomplete and unreliable when reporting the number of victims and the number of investigations that take place. The majority of states rely on the National Human Trafficking Resource Center (NHTRC) which tracks the number of telephone calls regarding the victimization associated with human trafficking because the states lack their own databases on a localized level. The NHTRC has demonstrated a steady increase of calls pertaining to human trafficking and exploitation which demands the need for more databases that could capture local issues that are not being reported or are severely underreported (Farrell & Reichert, 2017).

The Uniform Crime Reporting (UCR) Program was first introduced to track criminal activity in the 1920s, but human trafficking was not added to the database until 2013 and only thirteen states reported the crime to the database during that year. The number of states would increase to twenty-nine participating states in the next year and would slowly increase to encompass each state in the years that followed. The UCR represents approximately 95 percent of the agencies in the United States regardless of the jurisdiction (Farrell & Reichert, 2017).

The Summary Reporting System (SRS) and National Incident-Based Reporting System (NIBRS) are the main reporters to the UCR which includes the classification of the crimes. Part I type crimes involve serious offenses while Part II type crimes involve less common or less serious offenses. Through the use of the SRS and NIBRS, the crimes can be classified by where they were committed, perpetrator and victim characteristics, and if an arrest was made (Farrell & Reichert, 2017).

According to Farrell and Reichert (2017), the UCR is used more than any other database for research, allocation of resources, private entities, budgeting, and the media. While it is fortunate that the UCR is a national database that can provide the most detail on criminal activity, the database is only as good as the information that is provided. Many states do not report annually. Some do not report all of their crimes which makes the UCR database a victim of error by human and legislative fault (Farrell & Reichert, 2017).

While the United States battles human trafficking on a domestic level, there are many immigrants that enter the country who are victims of the crime. Horowitz (2019) stated that a T Visa may be issued to assist foreign people who have been smuggled into the country through means of human trafficking. A nineteen year old, Maria Choz from Guatemala, was the first person to obtain a T Visa from Attorney general Ashcroft in 2002. Unfortunately, as the country states the desire to assist immigrants who are victims of human trafficking, T Visas are not matching the number of requests each year which supports the notion that human trafficking victims are still isolated and have an unsupportive network of false hope. Horowitz (2019) stated that the T Visa's maximum grant of 5,000 per year was supposed to be enforced after 10 years of the first was granted yet the focus was on the prosecution of traffickers first and the victim protection would be an afterthought. While the Department of Justice (DOJ) states that between fourteen and eighteen thousand individuals are trafficked into the United States annually, the T Visa falls short of its intention (Horowitz, 2019).

The Dark Net

The creation of cyberspace has led to countless improvements to the world yet also allowed for different alternatives for criminals to use in their offenses. As technology changes constantly, criminals are able to adapt and broaden their attacks as they hone their skills of

evading law authorities and improving their financial gain and power. The introduction of the internet and cyberspace has led to the Dark Web which assists in anonymity and techniques for criminals to evade detection from law authorities (Portnoff, 2018).

The Dark Web, or Dark Net, is a network where people can anonymously venture through a level of cyberspace and partake in different activities that may be legal or illegal. Governments use the anonymity and encryptions which is the positive aspect of cyberspace like the Dark Net, however, criminals use it to hide their offenses from law authorities. Arguably, the first official Dark Net was ARPANET which was developed in 1969 for the use of the Pentagon to secure data through different computers that interacted with each other. ARPANET was used in the first known online transaction for a drug sale in 1971 by students attending Stanford University. After the Silk Road, an online drug bazaar, made public news broadcasts after whistleblowing occurred, the Dark Web became a subject in the public media (Omar & Ibrahim, 2020).

The Dark Net assumes that a product that is widely distributed is available to limited people, users could copy these products if they find it interesting and have the ability to copy, and all users are connected through channels of high bandwidth. There are many individuals that do not believe the Dark Net is as dangerous as others proclaim it is but it can be very dangerous when used by criminals or individuals who have illicit tendencies (Omar & Ibrahim, 2020).

The establishment of the internet led to the creation, accessibility, ability to share and trade child pornographic images with ease. Prior to cyberspace, these crimes were extremely difficult for criminals to commit since the access to consumers, producers, and distributors was limited. Cyberspace changes every day and now there are online human trafficking communities instead of an isolated offender. BBC found that there were several Facebook groups designated

for the trafficking of minors which illustrated the need for traceability, tracking, and prosecuting of these individuals however, the Dark Web has led to the inability to do so (Reid & Fox, 2020).

Reid and Fox (2020) discussed the case of Peter Scully who was discovered and arrested by Philippine investigators. Scully produced and broadcasted videos of sadistic abuse to children to an international ring of pedophiles on the Dark Net. In a video series called *Daisy's Destruction*, Scully raped, tortured, and he was perceived to have killed an eighteen month old child named Daisy. Scully also filmed the torture, rape and apparent murders of other children of various ages. It would later be discovered that Daisy sustained multiple physical injuries, a twelve year old who also appeared to be murdered was found alive, and an eleven year old was killed and buried in an area that Scully rented. The horrendous episodes of *Daisy's Destruction* paid Scully ten thousand dollars per view as the viewings were preferred by other sadistic criminals watching on the Dark Net (Reid & Fox, 2020).

The Onion Router (TOR)

The Onion Router (TOR), an anonymous network, was developed by the United States Naval Research Laboratory in 2002 but the public would gain access in 2004 and create a virtually untraceable atmosphere in the Dark Web which leads to the inability to track criminals. TOR blends cyber traffic with various international servers which creates a cloak of virtual invisibility. While TOR was developed to protect the country and its citizens, many terrorists have adopted its abilities and used it to their own political agendas of destruction. Approximately one and a half percent of Dark Web users use TOR for illicit purposes but the amount of traffic that transpires, most of it involving child abuse and child pornography, make up a great portion of the Dark Web (Omar & Ibrahim, 2020).

TOR is a virtual set of tunnels that does not fall under any jurisdiction, boundary, or authority to date. TOR is made up of multiple layers of encryptions which is why it efficient in its goals. People who use TOR on the Dark Web will be able to surf without fears of censorship or prosecution because there is no way to trace their movements in that area of cyberspace. When someone uses TOR, the online services can determine that TOR was used but there is no way to investigate the individual because the privacy TOR protects against it (Omar & Ibrahim, 2020).

Through the separation of routing and identification granted by TOR, an individual can surf the Dark Web, use chat functions, and engage in other types of internet activity without being tracked by other individuals including law enforcement (Omar & Ibrahim, 2020). TOR can protect against the analysis of cyber traffic, but it cannot protect the individual after an end-to-end correlation is performed which means someone else received an item such as an e-mail (Omar & Ibrahim, 2020). The Dark Net holds different types of child sexual abuse material (CSAM) that is viewed, bought, and sold through means of black markets in cyberspace and forums that are hosted TOR (Portnoff, 2018). TOR is a browser that allows users to explore the Dark Web anonymously by encrypting the networks that use it (Portnoff, 2018). Omar and Ibrahim (2020) stated that 80 percent of Dark Web traffic involves child pornography and abuse yet it is impossible to determine how much of the traffic is made up of criminals and how much is made up of law enforcement agencies because of the lack of traceability from TOR.

Black Market Forums in Cyberspace

A major issue of the Dark Net involves Black Market forums that are used as marketplaces and avenues of communication. The Dark Web is home to many underground areas where cybercrime flourishes through the exchange of trade relationships, information on stolen

personal identification such as credit card information, online credentials, and various compromised online hosts (Portnoff, 2018).

Cybersecurity professionals monitor forums, but they are also impacted by TOR anonymity and other barriers that the Dark Web provides. Law authorities have found monitoring labor-intensive which includes manpower, knowledge, training, financial ability, and more resources that the agency may not have to spare. These cyber black-market forums also have a rapid change in the vocabulary that is used to remain as untraceable as possible. Thousands of cybercriminals use black market forums to share CSAM, discuss how to groom children, and educate each other on how to avoid law enforcement (Portnoff, 2018).

Law Enforcement and the Dark Web

Law enforcement agencies have met many new challenges with the establishment of cyberspace and the Dark Web. Globalization and cyberspace have made traditional processes obsolete in many areas of TOCs especially when discussing the involvement of the Dark Web. In 2014, the World Drug Report discussed the Dark Web as a challenge to law authorities that was unique and continuously growing. In the same year, Interpol released the Internet Organized Crime Assessment (iOCTA) which explained that criminals do not need to be in the vicinity of the physical crime to engage in the activity and the Dark Web assisted in covering their tracks from a remote, undisclosed location (Omar & Ibrahim, 2020).

Rajasegaran (2017) stated that federal laws were established to prevent human trafficking starting with TVPA in 2000 yet the trafficking environment is constantly changing. Legislation must constantly change as new discoveries are made to prevent human trafficking from occurring, protecting and assisting the victims, and prosecuting the traffickers (Rajasegaran, 2017).

In 2011, the former U.S. Attorney General Eric Holder proclaimed that the internet had led to a steep rise in child pornography while the ages of the victims declined. Reid and Fox (2020) explained that over 80 percent of requests on the Dark Web was made up of child pornography which trumped other illicit activities such as gambling and drugs. Many criminals use cyberspace to conduct live broadcasts of exploitation persons through email, file sharing, websites, social networking, and instant messaging and have included the new feature of “Molestation on Demand” where people who are watching the live broadcasts can request specific actions as the abuse is already being conducted to fulfill their desires and fetishes (Reid & Fox, 2020).

The people who support child sex tourism schemes pay for the exploitation of children in other, lesser-developed countries. For example, the Peoples Recovery Empowerment Development Assistance (PREDA) estimated that thousands of children in the Philippines are exploited every year for the pleasure of wealthy people in other countries making requests of sexual abuse. In the Clear Net, social networking sites and online businesses often promise false hope and financial gain as they recruit, groom, and exploit the victims. Three-fourths of the minors who are victims of human trafficking were recruited through the use of the internet (Reid & Fox, 2020).

Additionally, mail-order brides are now being advertised on the internet yet the U.S. Citizen and Immigration Services (USCIS) states of almost nine thousand approvals where the ordered bride was a minor because a loophole of Visa sponsorships into the United States. Furthermore, labor trafficking has also increased because the establishment of the Internet where potential workers find work in different countries where they may not know much or any of the native language and sign agreements that they do not understand. These individuals may also be

charged recruitment fees, Visa renewal fees, and other fees that could exceed eight thousand dollars which adds to the helplessness and hopelessness of the trafficked individuals (Reid & Fox, 2020).

In 2015, a security threat released by IBM stated that more cyberattacks were being committed with the Dark Web and TOR networks which posed more issues for counter-terrorism agencies, governments, and security programs. Law authorities may be able to use MEMEX which is a software that categorizes websites used on the Dark Web. While this is a suggestion, it also removes freedoms from individuals who are not part of illicit activities and only use the Dark Web because their government is strict on their personal and cyber-related freedoms. The goal of law authorities is to prevent crimes from occurring and assisting the victims after they have been abused (Omar & Ibrahim, 2020).

The United States government stated that it wants to prosecute the criminals but the Dark Web should still be a place where law-abiding individuals can still remain anonymous. This goal may be possible if specific users of the Dark Web are targeted with a hacking tool to trace IP addresses such as those related to Playpen, a website dedicated to child abuse. Between the years of 2014 and 2015, the number of users on Playpen grew from 60,000 to 215,000 accounts and approximately 11,000 random individuals used Playpen per week. As the FBI used their hacking tool, they were able to capture information of regular users of Playpen as opposed to those who used TOR (Omar & Ibrahim, 2020).

Unfortunately, the Dark Web is virtually untraceable and the activation, validation, and normalization of child pornography is a fast-growing issue. The Dark Web has black market forums where human trafficking and abuse are normal and supported through a numeric rating

system. This rating system could include information and reviews based on physical appearance, friendliness, and sexual performance (Reid & Fox, 2020).

It is understood that cyberspace has led to easier routes that criminals use to traffic individuals yet the Dark Web is rising in their repertoire (Reid & Fox, 2020). Human trafficking is growing with the anonymity of the Dark Net and law enforcement is struggling to identify each issue such as classified advertisements, black market forums, and other means of criminal activity and interaction (Portnoff, 2018).

Artificial Intelligence (AI) and Algorithms

A decade ago, AI was unable to be used regarding human trafficking but with technological advances, it now has the ability to identify patterns quicker than ever (Wu, 2020). Wu (2020) explained that it takes seconds for AI and algorithms to identify patterns which may substantially be faster than a human mind can process in their lifetime. AI can assist law authorities by conducting searches and obtaining results up to 60 percent faster than a human being (Brody, 2019). However, Wu (2020) cautioned that there still must be human beings who oversee the AI who is equipped with common sense, effective reasoning, good judgement, and good intentions to make this path successful.

There are now mobile applications that are being developed and enhanced every day and some are available on mobile devices such as smartphones. These apps are able to be used when a Quick Response (QR) Code is scanned from a smartphone in public restrooms and other areas where victims may be able to find them. When the QR Code is scanned and law authorities are notified, the police officers can immediately request cellular phone records and determine if there is trafficking involved which could save countless lives and resources (Wu, 2020).

McGill University and Carnegie Mellon University united their efforts to use an algorithm called *InfoShield* to identify similarities and patterns in online advertisements. Criminal justice officials may be able to use this algorithm in the future to combat human trafficking in the Clear Net and the Dark Web. The universities are working together to mold these algorithms into more effective solutions because a severe limitation is that the algorithm will identify all similarities which may be misleading or take up too many resources to research each finding in cases. Still, many individuals are hopeful of *InfoShield* because it can be used to narrow down organized groups who are involved in human trafficking. *InfoShield* was used to accurately identify approximately 85 percent of advertisements that were connected to human trafficking (EurekAlert!, 2021).

Traffick Jam is another tool that relies on AI and algorithms to investigate human trafficking. *Traffick Jam* was created by Emily Kennedy at Carnegie Mellon University who had discussed various stressors with detectives regarding human trafficking and took action to alleviate these issues. *Traffick Jam* can be used to find advertisements and where these advertisements were posted in a map view based on cellular phone numbers. In 2019, *Traffick Jam* was used to successfully make an impact on over 600 criminal cases regarding human trafficking (Brody, 2019). Traffic Jam led to another tool called *Spotlight* which was introduced by celebrities Ashton Kutcher and Demi Moore in 2014. In 2019, 8,000 officers used Spotlight to investigate human trafficking crimes (Brody, 2019).

Before *Backpage* was taken down, *Traffick Jam* found that over 130,000 new sexual trafficking advertisements were posted daily. Unfortunately, less than a year later, the number rose to over \$146,000 daily on other sites which is why algorithms are essential to combatting human trafficking and saving victims. Since criminals constantly change their own patterns when

they discover what is being used against them, the AI and algorithms that are used now must constantly evolve to keep up or surpass them (Brody, 2019).

In 2014, DARPA released *Memex* which is used to develop tools associated with searches and analytics to support the efforts against human trafficking (Brody, 2019). *Memex* has already been used in New York as it sentenced Benjamin Gaston to 50 years to life for sexually abusing and holding a woman captive (Greenemeier, 2015). To date, *Memex* is used to monitor advertisements on the internet that are used to lure and trap people into a life of human trafficking (Greenemeier, 2015). *Memex* is helpful because about 90 percent of these advertisements are located on the dark web that are not monitored by mainstream search engines (Greenemeier, 2015). In 2017, the 70 million dollar *Memex* effort ended but the rewards were priceless as these tools were used to assist the National Center for Missing and Exploited Children (NCMEC) and law enforcement agencies (Brody, 2019).

TellFinder shows similarities to *Traffick Jam* but can search through the dark web and searches through faces and written advertisements to find missing persons and combat human traffickers. *TellFinder* has been used to investigate over 300 human trafficking cases a year instead of the 30 of one law enforcement agency and led to approximately 7,500 tips to the NCMEC in one year. By using key words and phrases and searching hobby or review boards on the dark web, *TellFinder* has assisted law enforcement authorities vastly in their war against human traffickers (Brody, 2019).

The *Domain Insight Graph (DIG)* tool was developed by researchers at the University of Southern California with the focus on graphics, charts, software, AI, and a search engine. Over 200 law enforcement agencies use the *DIG* and it is being discussed as a potential tool in the Department of Defense (DoD). The *DIG* uses cluster queries to determine if there is

advertisements that use obscure data such as odd or excessive uses of special characters or punctuation or a pattern of phone numbers for various advertisements. The *DIG* is based on a user-friendly environment which alleviates the issues of training on specific areas and can be altered by different investigators for specific domains (Cameron, 2021).

Moving forward, NCMEC has started to use *Traffick Cam* which was developed by Jane Quinn, Molly Hackett, and Kimberly Ritter in St. Louis. *Traffick Cam* is used to study accessories in a room, furniture, and carpets to indicate which hotel was used to photograph the victim. By identifying differences between different furniture or other décor used by different hotel chains, *Traffick Cam* can determine which hotel was used for the photography in an advertisement. Furthermore, *Traffick Cam* can be updated by virtually anyone who takes pictures of their hotel room and uploads to the database through a *Traffick Cam* mobile application (Brody, 2019).

XIX was also introduced by Emil Mikhailov in San Francisco to distinguish people in blurry photographs, security cameras, and mirrors and has a success rate of almost 100 percent. *XIX* can tag objects such as license plate numbers, specific people, words, and phrases that could assist in finding victims and traffickers (Brody, 2019).

Flipping the script, there are also AI tools that use bots to catch the traffickers through a series of text messages regarding price, meeting spots, services, and fetishes. This tool is called *Seattle Against Slavery* and was developed by Robert Beiser who developed an algorithm that law enforcement agencies can use to alter at their own discretion by making the bot a certain age, race, and if the bot is experienced or not. When law enforcement views the interaction between the bot and the criminal, they can decide whether they have enough information to prosecute the

person. In 2017, more than ten police departments had adopted the bot which had disturbed approximately eighteen thousand discussions (Brody, 2019).

One of the more well-known AI platforms is *Spotlight* which was developed in 2012 by Thorn, a company cofounded by celebrities: Ashton Kutcher and Demi Moore (Thorn, 2021). Kutcher and Moore joined forces with Thorn's Chief Executive Officer (CEO), Julie Cordua, to combat human trafficking after watching a video and learning that it is a serious issue the United States as well as the rest of the world. *Spotlight* was developed to increase data sharing, collaboration, and bypass bureaucratic structures that make it difficult to help victims of human trafficking (Thorn, 2021). Now, Thorn is able to assist people through "1) accelerating victim identification, 2) equipping platforms, and 3) empowering the public" through the technological developments of *Spotlight* (Thorn, 2021).

Research Design and Procedures

There are various studies that relate to the Dark Web and human trafficking. This section will discuss studies based on classified advertisements, surveys, interviews, and information that is available to the public.

Classified Advertisement Studies

Portnoff (2018) also studied the classified advertisement website called *Backpage* that is a popular area for human traffickers to advertise and sell their victims. *Backpage* was established in 2004 and is available to any person who has access to an email address. As stated, the Dark Net is used legally by many individuals and *Backpage* is used for the sell or rental of other items such as furniture yet the focus of this study was based on adult content. The researcher stated that approximately 80 percent of advertisements for sexual products including individuals are found on the Dark Net (Portnoff, 2018).

The human traffickers protect themselves by not writing of coercion of the trafficked individual; Instead, the traffickers word the advertisements as if they victim were writing the information about themselves. Furthermore, in 2015, MasterCard and Visa prohibited the use of their services on *Backpage* transactions for adult listings which shifted the acceptable currency on the Dark Net to cryptocurrency such as Bitcoin that can be used in untraceable manners. In 2017, adult listings were prohibited altogether on *Backpage* yet there are other areas that traffickers commit crimes on the Dark Net. The study was focused on any information that was obtainable before *Backpage* prohibited the use of their website for adult listings (Portnoff, 2018).

Prior to the prohibition, *Backpage* was home to over one million classified advertisements between January 2008 and September 2014. Portnoff (2018) classified the authors by individual telephone numbers and email addresses to determine that of the amount of advertisements they found, there were fewer than 400,000 individual authors. When the researcher studied the timeframe between December 11, 2016 and January 9, 2017, they found that classified adult advertisements distributed from the United States reached over 700,000 advertisements and more than 140,000 authors (Portnoff, 2018).

Surveys and Interviews

Portnoff (2018) stated that most of the research pertaining to human trafficking is based on surveys and interviews where the majority of individuals were advertised and sold in some type of cyberspace. In one survey, over 60 percent of human trafficking survivors were advertised online and almost half of the surveyed stated that *Backpage* was the website used. *Facebook* and *Craigslist* were also among the most used cyber platforms for the advertisement and selling of individuals. The survivors of human trafficking often find themselves interviewed

by various organizations and may have agreed to follow-up interviews for additional analyses (Portnoff, 2018).

Farrell and Reichert (2017) also used surveys in their study but their target included managers of the UCR and other contacts from the Association of State Uniform Crime Reporting Programs (ASUCRP). These researchers surveyed these individuals because states submit data to the FBI that pertain to specific information. The authors wanted to determine what information was received through reports of human trafficking and followed up with telephone or email reminders which results in 78 percent of respondents to their survey between the months of July and August in 2015. To fill in 22 percent gap of those who did not respond, the authors used information from the 2014 “Annual Report Crime in the United States” that was distributed by the FBI in September of 2015 (Farrell & Reichert, 2017).

The surveys displayed that over 14,000 law enforcement agencies were represented by the UCR which approximately 272 agencies in each state. The results of the surveys indicated that 86 percent of these law enforcement agencies reported their cases to the UCR program of their state and 70 percent of UCR managers indicated their state uses NIBRS. Of the responses, it was noted that ten states reported the arrest or offense of human trafficking to NIBRS and five states that did not report to NIBRS reported this information to the state UCR (Farrell & Reichert, 2017).

65 percent of UCR managers reported human trafficking as a requirement to be integrated into the state reporting yet a quarter of the responders did not. The respondents also were asked if the reporting of human trafficking was encouraged which results in almost 50 percent affirmation, over 20 percent stated that they did not, and almost 30 percent were unsure. Furthermore, only 12 percent of UCR managers responded that law enforcement authorities

could identify when human trafficking was occurring. The lack of consistency of reporting that was found in this study indicated that law enforcement agencies are not required to report to the same area which means the data will always be skewed. Farrell and Reichert (2017) then conducted case studies of Illinois and Tennessee to study the processes of data collection on human trafficking offenses.

The case studies also involved interviews of investigators and crime-reporting staff who had experience in human trafficking cases. The researchers included questions regarding the identification of human trafficking, the investigations, challenge, and verification strategies to determine if the case applied to human trafficking. These surveys and interviews indicated the following challenges in the reporting of human trafficking: training, identification of offenses, investigative support, and accurate reporting of human trafficking (Farrell & Reichert, 2017).

Portnoff's (2018) study started with questions presented to law enforcement authorities where it was discovered that law enforcement lacked tools to assist in preventing human trafficking, prosecuting human traffickers, and protecting the victims of human trafficking. The tools that the researcher focused on included ones to determine who is responsible for the production of CSAM products and abusing the children and tools that would determine who was purchasing the items or individuals. The researcher stated that tools were needed in these areas to save invaluable resources in the protection of human trafficking victims and the prosecution of individuals who are guilty of trafficking individuals (Portnoff, 2018).

Portnoff (2018) determined that automatic techniques such as anomaly detective devices could present a solution for an issue that has plagued law enforcement agencies since the establishment of cyberspace. Principal Component Analysis (PCA) was presented as a way to reduce the dimensions of an area and to assist in the tracking of network anomalies. By using

PCA techniques, law enforcement may be able to analyze user behaviors yet it was noted that it was not usually used for these actions (Portnoff, 2018).

Public Knowledge

Sutanrikulu, et al. (2020) established a study of socioeconomic factors that could influence the levels of cybercrime in different countries. The authors used the Internet Archive with research from Gwen Branwen to collect data of 89 markets on the Dark Web and more than 37 black market forums. The goal of the study was to determine if individuals who had a better understanding of technology were more prone to be human traffickers in the Dark Web (Sutanrikulu, et al., 2020).

The researchers were focused on any product offers that were evident on the Dark Web through names of items and vendors and where the products and vendors were located by country. After deleting duplication information, Sutanrikulu, et al., (2020) found data in 38 different markets between 2013 and 2015. The researchers found that the Dark Net is unreliable when determining where products and vendors are located because of the ease to present false information within the platforms which indicated limitations in their study to determine if the socioeconomic standing of an individual had severe effects on if it was a factor in users of the Dark Net (Sutanrikulu, et al., 2020).

Beasley (2018) chose the DHS because of the agency's dedication to human trafficking and immigrant legislation. Between October of 2017 and April of 2018, the DHS had thirty-seven individual materials to train first responders, law enforcement personnel, and the public on the dangers and issues of human trafficking. Beasley (2018), however, did not look into training materials for the private domain such as employees of DHS and other agencies. Additionally, Beasley (2018) only studied the materials that were presented to English speakers or readers and

could not compare if the material was consistent among different languages. The study focused on public knowledge from the DHS regarding the sex of individuals, coercion, victims, forced labor, modern-day slavery, sexual exploitation, forced criminality, and sexual abuse (Beasley, 2018).

Beasley (2018) found modern-day slavery was apparent in 32 percent of the public DHS materials and 84 percent of the materials presented sexual exploitation. Additionally, almost a quarter of the materials stated that males were perpetrators even if they were discussed in almost 70 percent of the materials altogether. Alternatively, females were discussed in over 70 percent of the materials where 16 percent were identified as perpetrators and 22 percent of females were discussed as victims (Beasley, 2018).

Additionally, Beasley (2018) indicated that almost 80 percent of the materials depicted minors. Coercion was presented in almost 90 percent of the materials and forced labor was presented in almost 80 percent of the materials. Victimhood was present in each material making it the only keyword that was presented at 100 percent of the DHS materials while forced criminality only appeared in three percent of the materials. The DHS also referred to modern-day slavery instead of human trafficking in 32 percent of the presented materials where it was defined as forced labor and sexual exploitation (Beasley, 2018).

Horowitz (2019) researched documentation pertaining to T Visas that was presented from the executive and legislative branches of the United States and interviewed representatives of law enforcement and legal clinics who assisted in human trafficking cases. The documents that were reviewed included the TVPA between the years of 2000 and 2015 as it was revised through the years to analyze the use of words within the act and its revisions (Horowitz, 2019).

In 2000, the TVPA was introduced and was questioned about where criminality would fall in terms of sexual acts such as prostitution as to whether the trafficker or the trafficked individual would both be charged or just the trafficker. Additionally, the T Visa did little to protect the victims because their citizenship status as illegal immigrants. The TVPA 2000 did state that law enforcement authorities should use caution when handling these cases and not punish the individuals who are being trafficked (Horowitz, 2019).

In 2001, the TVPA was heard at the House of Representatives and United States Senate to clarify any issues or confusion of the proposed act. Since sex was mentioned more than labor in hearings and the act, it was argued that females were demonized because they were trafficked for sexual acts. The TVPRA would be introduced as a revised version of the TVPA in 2003 which broadened the description of human trafficking victims to males, females, and minors on an international scale (Horowitz, 2019).

The TVPRA 2003 also indicated that issues would pertain to the illegal immigrant status of the victims into the United States regarding the T Visas. Victims were encouraged to assist law enforcement agencies regarding their cases for assistance in obtaining a T Visa and the Attorney General was required to provide metrics on applications, approvals, and denials of all T Visa traffic (Horowitz, 2019).

In 2005, the TVPRA was revised to include additional justification for female victims to represent the 80 percent of cases that were addressed each year. Within this revision, the T Visa was not mentioned in the amendment but additional funds were allowed for DHS investigations. Three years later, TVPRA 2008 would introduce the issues of trauma, T Visa extensions, and T Visa waivers of disqualification if the victim displayed bad moral character. The trauma extension declared that victims were not required to be interviewed by law enforcement if they

experienced trauma that was psychological or physical. This revision was aimed more at protecting the victims instead of assisting law authorities in prosecuting the human traffickers (Horowitz, 2019).

TVPRA 2011 extended the eligibility and protection of T Visas to family members of trafficked victims. In a Committee on the Judiciary meeting in 2011, it was questioned why T Visas were not issued to the maximum of their allotment and the answer was simply that victims distrusted law authorities and it was difficult to find victims in the United States even though it was also stated that the TVPRA was successful in its implementation. This discussion would lead to TVPRA 2013 that instructed law enforcement agencies of local, state, and federal levels collaborate more efficiently with victim service providers. The T Visa was discussed as an encouragement tactic for victims to assist law enforcement rather than its proposed purpose of protection for the victims (Horowitz, 2019).

Artificial Intelligence Research

Artificial Intelligence, referred to as AI, is a relatively new aspect of technological advances. While there are many studies regarding AI, there is very little in retrospect to the dark web and human trafficking, consecutively.

In one study, a form of AI called a learner was researched to determine if it could differentiate between unlabeled and labeled data regarding advertisements that law enforcement officials would be interested in. The test were supervised and semi-supervised at different times and the results from the test was verified by a human expert for accuracy. While the researchers did not compare their methodology with previous studies, they did draw comparisons from different learners, both supervised and semi-supervised (Alvari, et al., 2017).

In 2016, the researchers collected approximately twenty-thousand listings that were available to the public on *Backpage* and collected a time stamp, IDs, ages, titles, telephone numbers, descriptions, and locations. While this information was gathered, the study focused on the typed parts of the advertisements. The researchers compared data through methodology and previous approaches and evaluated data to determine if the learner was effective. The researchers concluded that the learner could be used to combat human trafficking and proposed that more research should be conducted in this area (Alvari, et al., 2017).

In a study of South Asia, Dutta (2020) studied AI and how it can be used to combat human trafficking among borders by using machine language or other tools to identify traffickers, routes, and/or faces of victims. Dutta also discussed how AI can be used to identify people after natural disasters who have been forced into trafficking which has plagued South Asia and other parts of the world. The research reported specific AI developments such as *Traffick Jam*, *Hotels50K*, and *Be My Protector* which have shown great promise through civilian inputs in the ability to track images from hotels to intervene in the criminal activities. Dutta reported approximately eight children are saved each day which is a rate of over 60 percent faster than previous results before AI. Unfortunately, many areas may not have sufficient funding to implement AI in their criminal justice systems which is the case in many Asian countries. The Fair Employment Agency (FEA) has assisted in these shortfalls but there still needs to be more research in the area (Dutta, 2020).

Summary

This literature review explained previous studies that pertain to human trafficking. The theoretical framework discussed the General Strain Theory, the Lifestyle Exposure Theory, and the Routine Activities Theory as ways to explain how human trafficking is accomplished and

how the criminal and victims are introduced in the world. Additionally, this section described different facets that must be understood prior to this study: human trafficking, the Dark Web, TOR, Black Market Forums, Legislation, AI, algorithms, and the challenges that Law Enforcement faces. Furthermore, this section described how various research designs and procedures were implemented in prior studies such as surveys, interviews, and the access of public databases. The next section will focus on the methodology, data collection processes, and analysis procedures for the current study.

CHAPTER THREE: METHODS

Overview

Human trafficking is not new to the world and legal action has been implemented to combat the crimes yet more is needed with the evolution of the internet causing issues that have not yet been fully scoped. This dissertation study consisted of a quantitative-comparative study regarding four of the AI platforms that have been mentioned in this study in a span of six years (2016-2021). The AI platforms that were compared include: *Spotlight*, *Traffick Jam*, *DIG*, and *Traffick Cam*. These platforms were chosen because they were developed and released before the start of 2016: *Spotlight* in 2014 (Farrell, 2016); *Traffick Jam* in 2013 (Waltz, 2017); *DIG* in 2016 (Cameron, 2016); and *Traffick Cam* in 2015 (Traffick Cam, n.d.).

The goal for this study was to determine if platforms were producing accurate and consistent results while tracking human trafficking as well as determine their effectiveness over the span of six years. It was preferable that AI platforms are gaining accuracy and consistency in their timelines. For example, if one platform shows continuous improvement since it was produced, the platform should be used more than one that shows little to no improvement or worse, a decrease in effectiveness.

Design

This was a quantitative-comparative study that compared four different AI platforms to determine how well they performed in combatting human trafficking crimes. This study analyzed the similarities and differences between the four AI platforms in the research questions to understand the data among each platform, individually and collectively (Richardson, 2018). Bettencourt and Lobo (2019) utilized a quantitative-comparative study when researching different cities such as how the societies had adapted to society in relation to others. The

researchers stated that the most important information to obtain through these comparisons is “how different societies deal with general problems affecting them all, including energy and resource extraction, and the organization of their socioeconomic networks over time” (Bettencourt & Lobo, 2019). The reasoning is similar to the purpose of this dissertation with the exception that the cities are AI platforms. While there are many different ways to battle human trafficking, the algorithms are becoming more essential because of the evolution of cyberspace including the dark web.

Research Questions

The following questions were posed regarding this study:

RQ1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

This research question is addressed in two separate ways. First, the researcher utilized different websites to collect data regarding the AI platforms’ performance regarding how many incidents were assisted, how many victims were assisted, and how many traffickers were convicted regarding *Spotlight*, *Traffick Jam*, and *DIG*. Secondly, *Traffick Cam* was assessed through with metrics pertaining to how many hotels were in the app, how many images were in the app, and how many downloads of the app occurred in comparison to global numbers.

RQ2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

This research focused on the differences in different AI platforms and how consistent their rates of assisting cases, helping victims, and convicting traffickers had been in the last six years. It also focused on how consistent they had been in comparison to the other AI platforms in

this study. This indicated if the platforms were remaining static, gaining accuracy, or were failing to remain as consistent as their previous years.

RQ3: *In a span of six years, how effective has detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

While AI platforms may incorporate the same standardized tools, it was investigated how effective they were as well. The definition of the effectiveness of the AI platforms depends on the individual as it could relate the protection of human beings and/or economic means (Wu, 2020). The student tried to compare at least three different websites per platform and obtain the metrics pertaining to how many cases were assisted, how many victims were assisted, how many traffickers were convicted, how many hotels and images of hotels were uploaded into the *Traffick Cam* app, and how many global downloads of *Traffick Cam* occurred.

Hypotheses

The first research question was used to determine the accuracy of each AI platform in this study.

RQ1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

The hypotheses are as follows:

H₀1: The accuracy of the following algorithms remained constant in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

H_a1: The accuracy of the following algorithms has improved in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

The second question will be used to determine the rate of consistency the platforms mentioned in this document in a span of six years.

RQ2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

The hypotheses are as follows:

H₀1: The consistency rate of the following human trafficking AI platforms has remained constant in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

H_a1: The consistency rate for following human trafficking AI platforms has improved in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

The third question will be used to determine the effectiveness of the platforms mentioned in this document in a span of six years.

RQ3: *In a span of six years, how effective has detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

The hypotheses are as follows:

H₀1: The effectiveness of the following human trafficking AI platforms has remained constant in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

H_a1: The effectiveness of the following human trafficking AI platforms have improved over a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

Participants and Settings

The participants in this study included the following AI platforms: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*. The researcher tried to use information based on a timeframe of six years (2016-2021) from the official website for each platform, a government website and/or another website that includes data that has been documented to reduce bias. If the researcher could not find enough data on each platform, it is noted in the results. The average of these rates per year were used in the overall research.

These platforms that were researched and discussed provide an understanding of how they were used and how effective they had been in a span of six years based on public knowledge to ensure that there were no leaks to the human traffickers who want to learn how to evade the criminal justice more than they already do. The algorithms must constantly change as criminals identify what they are searching for. This study did not give human traffickers the opportunity to utilize information that they could not find in the public domain.

Instrumentation

The instrument used for this study was a Microsoft Excel sheet that was used to document the information obtained from various websites. The researcher used as many as three sources to collect rates from the documented success of these platforms. These numbers were then averaged per year per AI platform. The platforms were then compared with each other to determine what platforms showed the best, worst, and static of rates of improvement in the last six years in relation to accuracy, consistency, and effectiveness.

Procedures

The data processing and analysis included gathering metrics from various AI platforms from online clear net web sources. The data will only include data within a six year period

(2016-2021). This information was collected on a Microsoft Excel spreadsheet and then transferred to a Google Document. The procedure was as follows:

1. Submitted the IRB application
2. IRB application was approved
3. Research began with researcher identifying different websites:
 - a. *Traffick Jam*
 - b. *DIG*
 - c. *Spotlight*
 - d. *Traffick Cam*
4. The researcher used a government website for each platform, if available, to reduce bias
5. The researcher collected data from websites as it was available to reduce bias
6. The researcher gathered the data and input the information in Microsoft Excel
7. The researcher determined the average of each platform's data rates from their associated websites
8. The researcher utilized Google Docs to create graphs from information obtained in research
9. The researcher presented results and a discussion based on the evidence and comparison with the hypotheses
10. The researcher defended the research to their peers

Data Analysis

Accuracy

The accuracy of this study was based on how accurate the AI platforms were between the sources that were compared individually and in comparison to larger areas of interest. For example, Spotlight sources were compared to determine if the public webpages were projecting the same, or similar, metrics to determine if Spotlight was accurate, or mostly accurate, across the findings. The average of these findings were then taken and compared to international data to determine if Spotlight had made an impact on assisting victims, assisting in cases, and/or convicting traffickers. Likewise, the other AI platforms (*DIG*, *Traffick Jam*, and *Traffick Cam*) were also compared in this way to determine accuracy of each.

Consistency

The study of consistency was completed by collecting and comparing data using the platform(s) that are publicly available. The consistency results reflected that there is a lack of consistency in each AI platform and it was determined that information was inconclusive. This study found that *Spotlight* was the only AI platform that could truly be compared with a larger scale regarding different international publications. While *Traffick Cam* did show information regarding global numbers, the researcher did not find other global sources to compare with each other in comparison to *Traffick Cam*. Therefore, *Spotlight* was the only AI platform utilized in this question.

Effectiveness

The effectiveness of the platforms was determined with the information provided by the accuracy and consistency of this dissertation to determine if rates had changed among the AI

platforms in a span of six years and if they were truly effective in the real-world. While each AI platform was compared in this area, the results were the same results of accuracy. There is a difference between the two: while accuracy questioned if the information is accurate among the research, the effectiveness asked if these AI platforms made a viable difference in the international publications they were compared to.

Accuracy and Bias

It was important to note that all studies have bias and some information may be present while other information may not be. Furthermore, one website may have had different information than another which could have skewed the results and produced outliers. To decrease the chance of bias, the researcher attempted to gather at least three sources per platform and gather the average amount per year of the study (2016-2021). By doing so, the average would help lessen the amount of inaccurate data if several websites have different metrics. When possible, the researcher utilized the platform's official website, a government website, and another website as needed to construct the best results for this study (POCD, n.d.).

Summary

This section indicated the methodology that was used to provide a quantitative-comparative study through the use of public knowledge of various AI platforms in a span of six years (2016-2021). The information was gathered from various websites pertaining to *Traffic Jam*, *DIG*, *Spotlight*, and *Traffic Cam* and used in a Microsoft Excel sheet for sorting and exporting to Google Docs to present illustrative data. This process was used to assist in determining how accurate, consistent, and effective algorithms were in a span of six years.

CHAPTER FOUR: FINDINGS

Overview

The purpose of this study was to determine if the *Spotlight*, *Traffick Jam*, *Traffick Cam*, and/or *DIG* AI platforms had evidence that support their programs are accurate, consistent, and/or effective in the battle against human trafficking. Additionally, this study was to determine if the same AI platforms had decreased in their abilities to produce accurate, consistent, and effective results, or if they had remained constant in the years of 2016, 2017, 2018, 2019, 2020, and 2021. The information was obtained from online web pages with specific search criteria to meet each year of the study. Appendix B denotes where each piece of information was derived from.

This chapter displays the data that was recovered from this quantitative-comparative study in the format of tables and graphs that help support or reject the null and alternate hypotheses for the following research questions:

RQ1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

- H₀₁: The accuracy of the following algorithms remained constant in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*
- H_{a1}: The accuracy of the following algorithms has improved in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

RQ2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

- H₀₁: The consistency rate of the following human trafficking AI platforms has remained constant in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.
- H_{a1}: The consistency rate for following human trafficking AI platforms has improved in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

RQ3: *In a span of six years, how effective has detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

- H₀₁: The effectiveness of the following human trafficking AI platforms has remained constant in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.
- H_{a1}: The effectiveness of the following human trafficking AI platforms have improved over a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

The results that were obtained in this study are defined further in the following sections of this chapter. The descriptive statistics indicates how the research questions were divided and provides the tables and figures to support the arguments pertaining to the accuracy, consistency, and effectiveness of each AI platform in this study.

Descriptive Statistics

The research questions were divided in specified areas depending on what the researcher found is reported on each. RQ1 was divided into two separate parts because *Traffick Cam* did not present the same type of data found in *DIG*, *Spotlight*, and *Traffick Jam*. Due to the nature of the data found in *Traffick Cam*, *Traffick Cam* was only used in RQ1b of this research question.

RQ1a was used to study the accuracy using *DIG*, *Spotlight* and *Traffick Jam* data while RQ1b only used information regarding *Traffick Cam* in relation to accuracy. This division was made because *DIG*, *Spotlight*, and *Traffick Jam* indicated information pertaining to the human trafficking incidents/cases assisted, the victims that were assisted, and the human traffickers that were convicted while information regarding *Traffick Cam* did not include these metrics. *Traffick Cam* information revolved around how many hotels were in the app's directory, how many images of hotel rooms were uploaded to the app, and how many users around the world had downloaded the app. Because the metrics did not imply similar information for *Traffick Cam* in comparison to *DIG*, *Spotlight*, and *Traffick Jam*, the research question was divided into parts RQ1a and RQ1b. RQ2 focused on the consistency but *Spotlight* was the only AI platform that was compared due to a lack of data found that could be compared to other sources such as the Trafficking in Persons (TIP) Reports and other nationwide statistical information. RQ3 focused on the effectiveness of each AI platform: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*. After each research question, the results are written but more in-depth information will be provided in the Results section of this chapter.

RQ1: Accuracy

This research question was addressed in two separate ways. *DIG*, *Spotlight*, and *Traffick Jam* were used in RQ1a while *Traffick Cam* was used in RQ1b. The researcher collected data from public web pages and compared the data with each other to determine what impact each AI platform had in comparison to national reports.

RQ1a: DIG

The AI platform, *DIG*, was developed to utilize cluster queries to determine if online advertisements were used to promote human trafficking and assist law enforcement agencies in

combatting the crimes associated with the TOC (Cameron, 2021). The results are presented in tables because there is very little data to support the argument of accuracy regarding DIG.

DIG 2016

No data to report

DIG 2016 vs. US 2016 TIP Report

Table 1 Accuracy: DIG Metrics vs TIP Report 2016

2016			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	2713	7956	439
DIG (Average)			

DIG 2017

Table 2 Accuracy: DIG Metrics 2017

	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
2017			
Source 1	3		
Source 2	3		

DIG 2017 vs. US TIP Report

Table 3 Accuracy: DIG Metrics vs TIP Report 2017

2017			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1,845	11,992	499
DIG (Average)	3		

DIG 2018

Table 4 Accuracy: DIG Metrics 2018

	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
2018			
Source 1	20		

DIG 2018 vs. US TIP Report

Table 5 Accuracy: Table 1 DIG Metrics vs TIP Report 2018

2018			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1773	12552	526

DIG (Average)	20		
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DIG 2019

No data to report

DIG 2019 vs. US TIP Report

Table 6 Accuracy: Table 1 DIG Metrics vs TIP Report 2019

2019			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1830	11976	475
DIG (Average)			

DIG 2020

No data to report

DIG 2020 vs. US TIP Report

Table 7 Accuracy: Table 1 DIG Metrics vs TIP Report 2020

2020			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1865	14571	309

DIG (Average)			
------------------	--	--	--

DIG 2021

No data to report

DIG 2021 vs. US TIP Report

No data to report

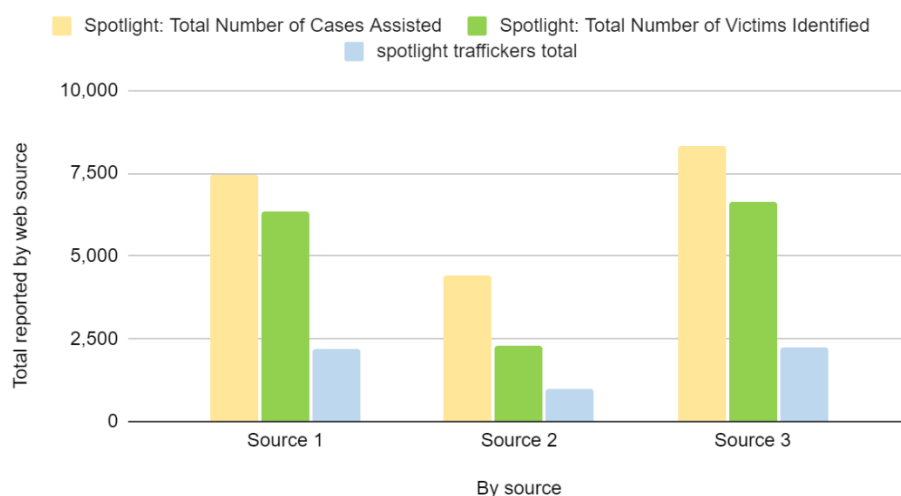
RQ1a: Spotlight

Spotlight is an AI platform that uses data sharing to assist in identifying victims of human trafficking and assisting the public in the fight against the TOC (Thorn, 2021). It may have more influence over other AI platforms because of its public awareness by individuals, namely celebrities, who have stake in the AI platform: Ashton Kutcher and Demi Moore. The researcher compared the metrics individually and with international TIP reports to determine if the websites were reporting consistent results with each other before comparing them to other types of reports. This assisted in determining if the Spotlight platform had outliers in data when comparing it to the amount of assistance it provided to law enforcement agencies and victims.

Spotlight 2016

Figure 1 Accuracy: Spotlight Metrics 2016

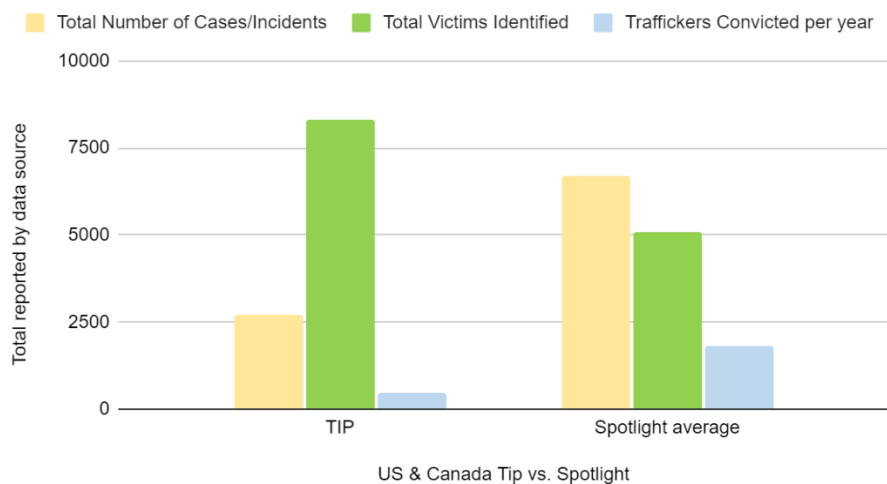
Spotlight: 2016



Spotlight 2016 vs. Combined 2016 TIP Reports (US and Canada)

Figure 2 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2016

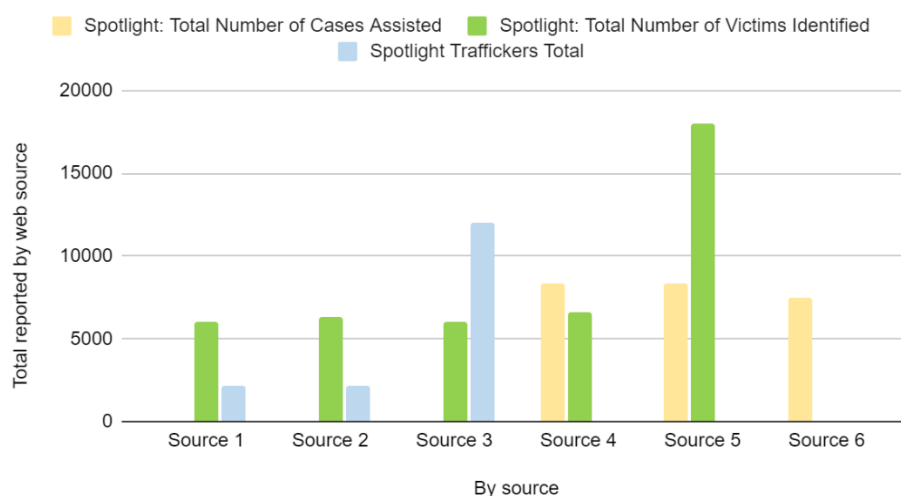
(2016) US & Canada TIP vs. Spotlight average



Spotlight 2017

Figure 3 Accuracy: Spotlight Metrics 2017

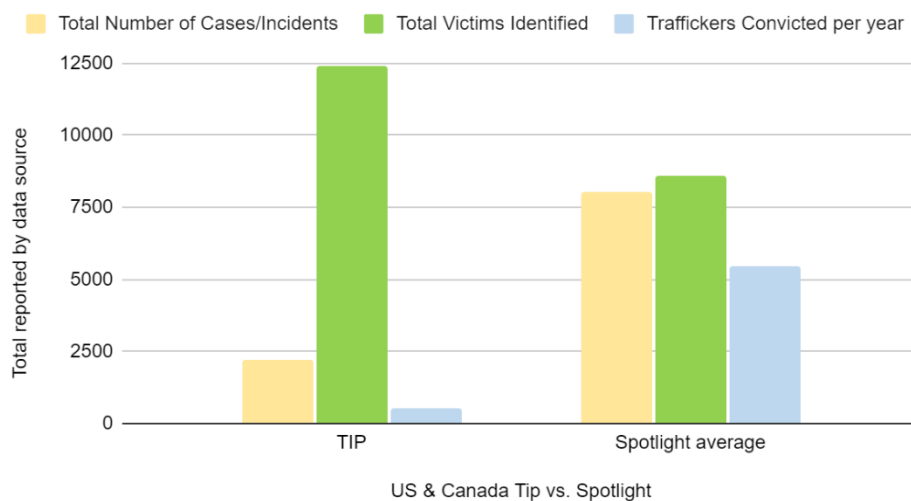
Spotlight: 2017



Spotlight 2017 vs. Combined 2017 TIP Reports (US and Canada)

Figure 4 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2017

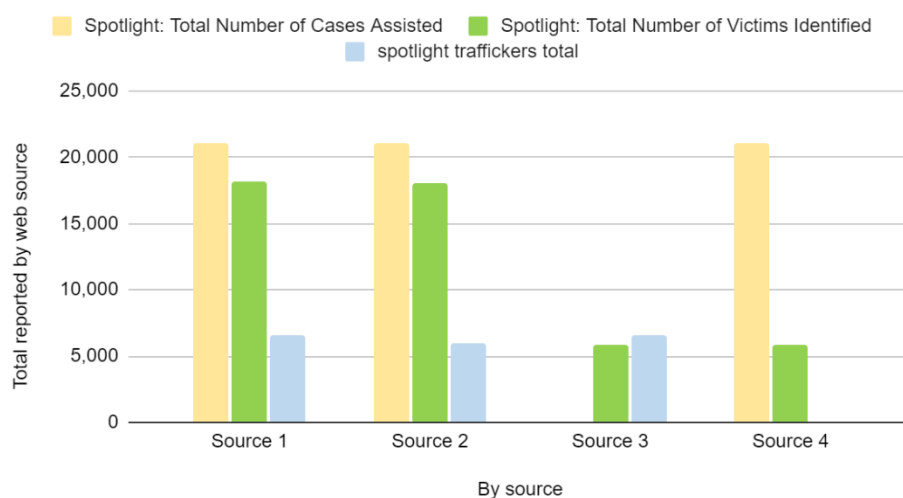
(2017) US & Canada TIP vs. Spotlight average



Spotlight 2018

Figure 5 Accuracy: Spotlight Metrics 2018

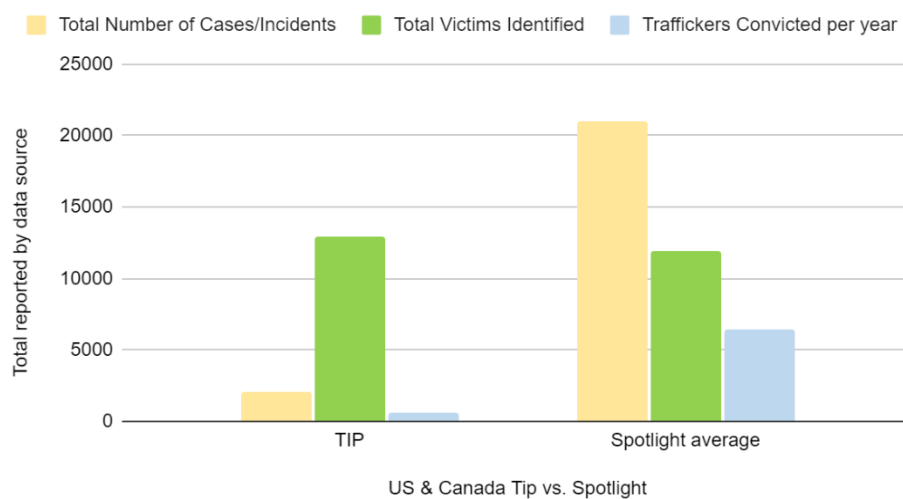
Spotlight: 2018



Spotlight 2018 vs. Combined 2018 TIP Reports (US and Canada)

Figure 6 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2018

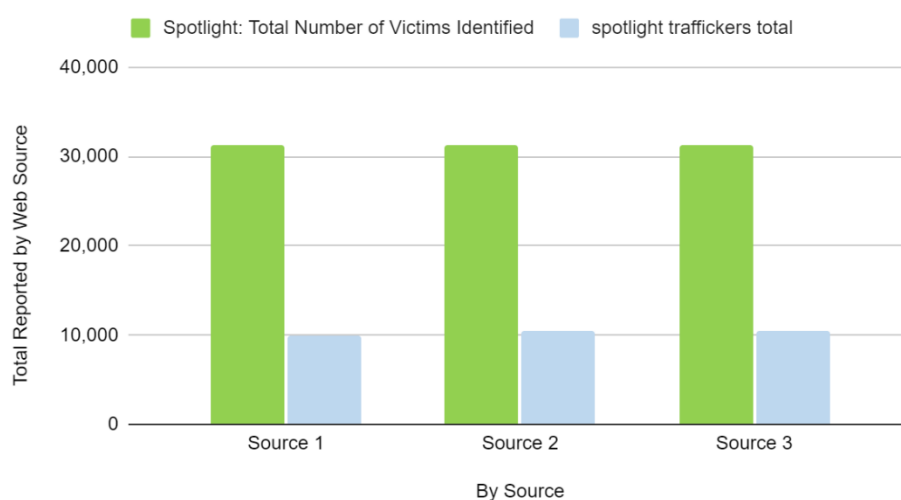
(2018) US & Canada TIP vs. Spotlight average



Spotlight 2019

Figure 7 Accuracy: Spotlight Metrics 2019

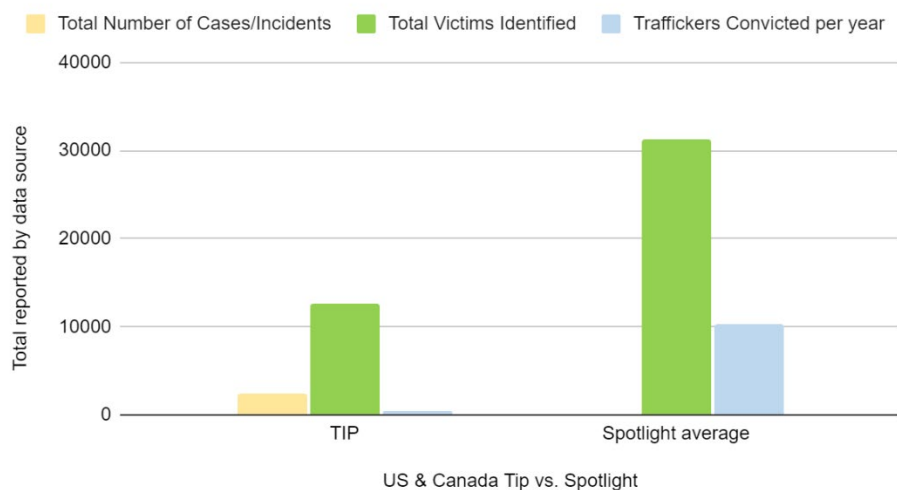
Spotlight: 2019



Spotlight 2019 vs. Combined 2019 TIP Reports (US and Canada)

Figure 8 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2019

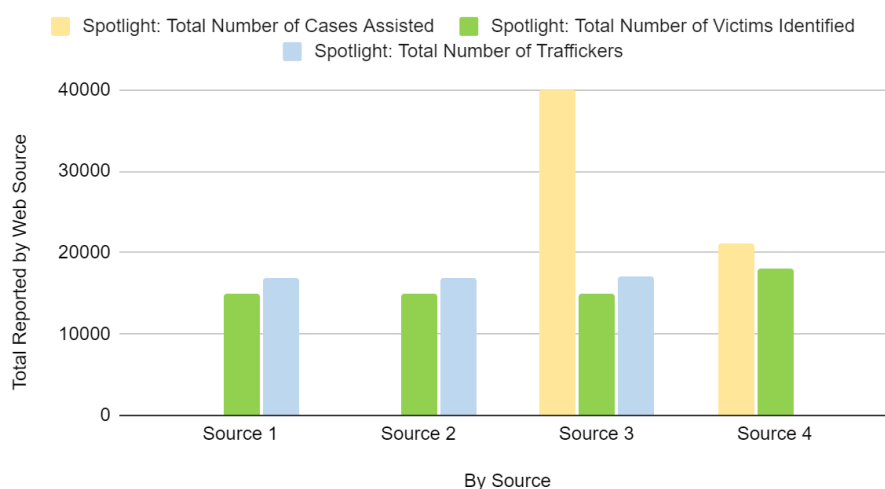
(2019) US & Canada TIP vs. Spotlight average



Spotlight 2020

Figure 9 Accuracy: Spotlight Metrics 2020

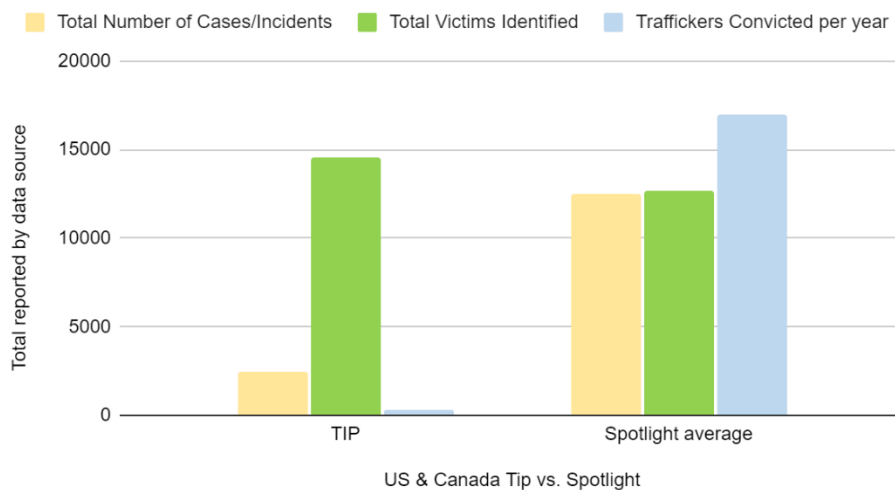
Spotlight: 2020



Spotlight 2020 vs. Combined 2020 TIP Reports (US and Canada)

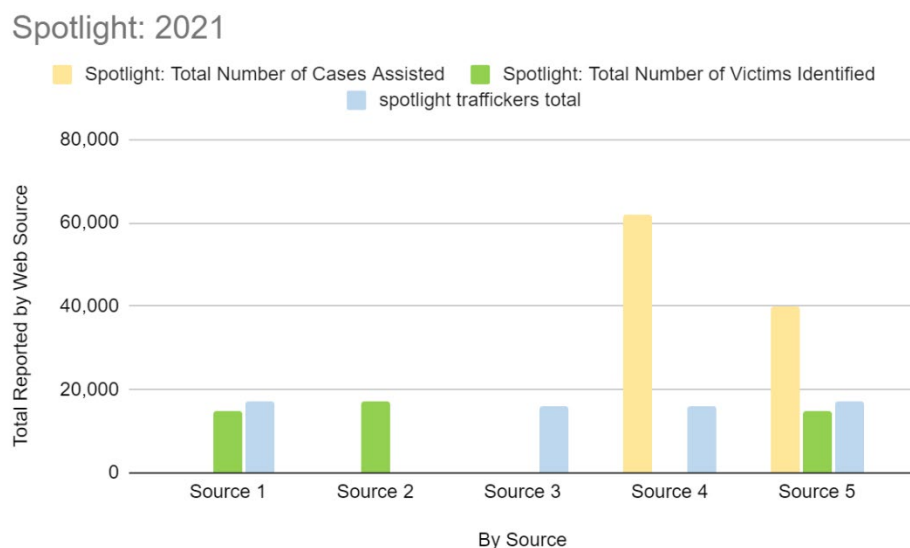
Figure 10 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2020

(2020) US & Canada TIP vs. Spotlight average



Spotlight 2021

Figure 11 Accuracy: Spotlight Metrics 2021



Spotlight 2021 vs. Combined 2021 TIP Reports (US and Canada)

Table 8 Accuracy: Spotlight Metrics vs combined US and Canada TIP Reports 2021

Combined US & Canada TIP vs. Spotlight			
2021	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	Not Available	Not Available	Not Available
Spotlight average	51,000	11,197	16,500

RQ1a: Traffick Jam

Traffick Jam is an AI platform produced by Emily Kennedy and Marinus Analytics and it is based on using cellular phone numbers and map views. The metrics were compared to three country's TIP reports where *Traffick Jam* is utilized.

Traffick Jam 2016

Table 9 Accuracy: Traffick Jam Metrics 2016

2016	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam traffickers total
Source 1		300	

Traffick Jam 2016 vs. US, Canada, and UK 2016 TIP

Table 10 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2016

2016			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	2,713	12,128	452
Traffick Jam		300	

Traffick Jam 2017

No data to report

Traffick Jam 2017 vs. US, Canada, and UK 2017 TIP

Table 11 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2017

2017			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year

TIP (US, UK & Canada)	2,220	17,550	508
Traffick Jam			

Traffick Jam 2018

Table 12 Accuracy: Traffick Jam Metrics 2018

2018	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Convicted Traffickers per Year
Source 1		3,000	

Traffick Jam 2018 vs. US, Canada, and UK 2018 TIP

Table 13 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2018

2018			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	3,237	19,965	582
Traffick Jam		3,000	

Traffick Jam 2019

Table 14 Accuracy: Traffick Jam Metrics 2019

2019	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Convicted Traffickers per Year
Source 1	600		
Source 2		3,800	

Traffick Jam 2019 vs. US, Canada, and UK 2019 TIP

Table 15 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2019

2019			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	3,820	23,179	531
Traffick Jam	600	3,800	

Traffick Jam 2020

Table 16 Accuracy: Traffick Jam Metrics 2020

2020	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Convicted Traffickers per Year
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Source			
1		3,800	
Source			
2		6,800	
Source			
3		6,800	

Traffick Jam 2020 vs. US, Canada, and UK 2020 TIP

Table 17 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2020

2020			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	4,283	25,184	316
Traffick Jam		5,800	

Traffick Jam 2021

Table 18 Accuracy: Traffick Jam Metrics 2021

	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Convicted Traffickers per Year
2021			
Source			
1		3,800	

Source 2		3,800	
Source 3		3,800	

Traffick Jam 2021 vs. US, Canada, and UK 2021 TIP

Table 19 Accuracy: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2021

2021			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	Not available	Not available	Not available
Traffick Jam		3,800	

RQ1a Indications: According to the data, *DIG* held extremely low numbers for the years that data was available and supports that the AI platform remained constant and null hypothesis was accepted which rejected the alternate hypothesis. The null hypothesis is accepted regarding *Traffick Jam* because the information also remained consistent. *Spotlight* does not remain constant and seems to improve meaning the null hypothesis is rejected and the alternate hypothesis is accepted. However, it should be noted that more complete datasets should be sought for future research to strengthen the hypotheses.

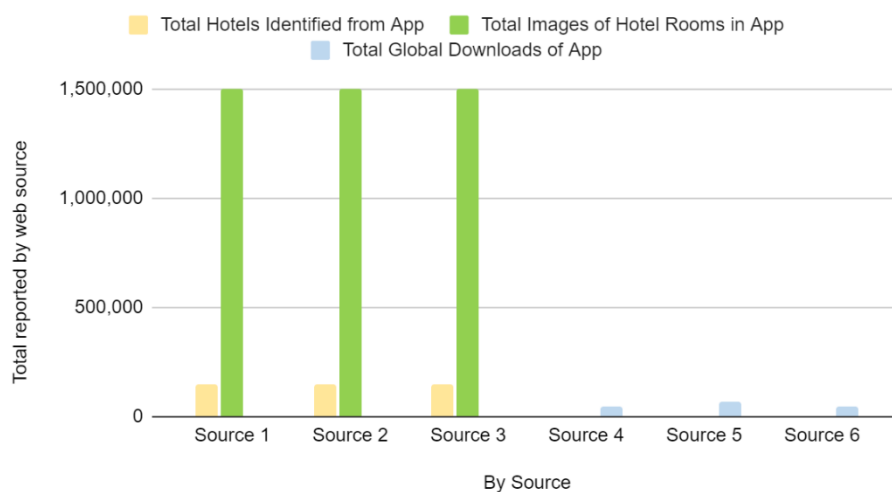
RQ1b: Traffick Cam

Traffick Cam is a mobile app that houses numerous hotels and images of hotel rooms to compare against images that could provide the location of a victim of human trafficking. The images and information can be uploaded by anyone who is venturing around the world to combat human trafficking (Brody, 2019). Again, the data was compared with each other and then compared to global numbers to indicate how accurate the metrics are in comparison to each other.

Traffick Cam 2016

Figure 12 Accuracy: Traffick Cam Metrics 2016

Traffick Cam (2016)



Traffick Cam 2016 vs. Global Hotel Numbers 2016

Table 20 Accuracy: Traffick Cam Metrics vs Global Metrics 2016

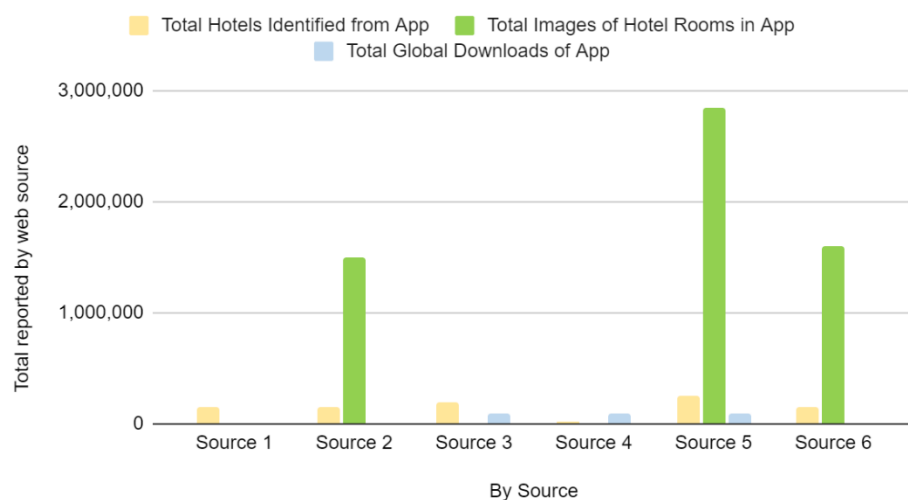
Total Global Numbers Compared to Traffick Cam App		
2016		
	Total Hotels	Total Global Downloads

Global Numbers	182,149	140.68 Billion
Traffick Cam Average	145,000	56,666

Traffick Cam 2017

Figure 13 Accuracy: Traffick Cam Metrics 2017

Traffick Cam (2017)



Traffick Cam 2017 vs. Global Hotel Numbers 2017

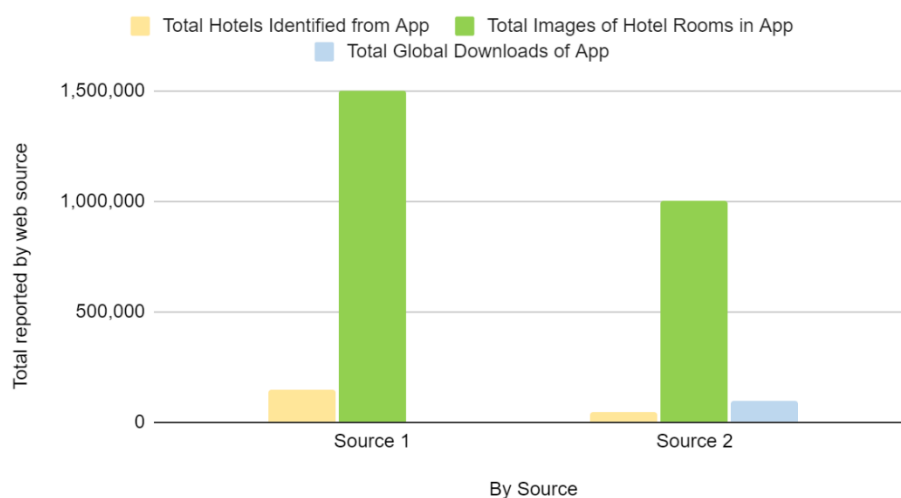
Table 21 Accuracy: Traffick Cam Metrics vs Global Metrics 2017

Total Global Numbers Compared to Traffick Cam App		
2017		
	Total Hotels	Total Global Downloads
Global Numbers		175 Billion
Traffick Cam Average	152,333	97,615

Traffick Cam 2018

Figure 14 Accuracy: Traffick Cam Metrics 2018

Traffick Cam (2018)



Traffick Cam 2018 vs. Global Hotel Numbers 2018

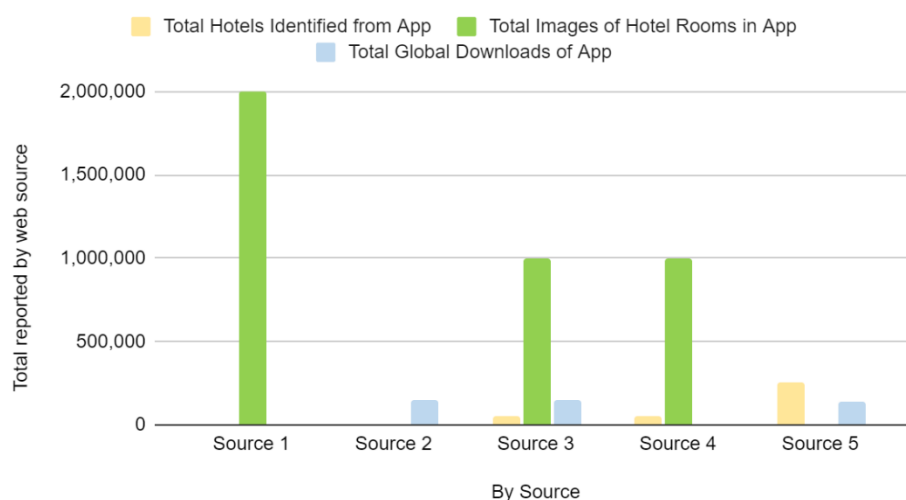
Table 22 Accuracy: Traffick Cam Metrics vs Global Metrics 2018

Total Global Numbers Compared to Traffick Cam App		
2018		
	Total Hotels	Total Global Downloads
Global Numbers	184,229	192.45 Billion
Traffick Cam Average	97,500	150,000

Traffick Cam 2019

Figure 15 Accuracy: Traffick Cam Metrics 2019

Traffick Cam (2019)



Traffick Cam 2019 vs. Global Hotel Numbers 2019

Table 23 Accuracy: Traffick Cam Metrics vs Global Metrics 2019

Total Global Numbers Compared to Traffick Cam App		
2019		
	Total Hotels	Total Global Downloads
Global Numbers	700,000	203 Billion
Traffick Cam Average	118,333	148,000

Traffick Cam 2020

Table 24 Accuracy: Traffick Cam Metrics 2020

	Total Hotels Identified	Total Images of Hotel Rooms	Total Global Downloads
2020	from App	in App	of App

Source			
1	500,000	3,000,000	

Traffick Cam 2020 vs. Global Hotel Numbers 2020

Table 25 Accuracy: Traffick Cam Metrics vs Global Metrics 2020

Total Global Numbers Compared to Traffick Cam App		
2020		
	Total Hotels	Total Global Downloads
Global Numbers	700,000	218 Billion
Traffick Cam Average	500,000	

Traffick Cam 2021

Table 26 Accuracy: Traffick Cam Metrics 2021

2021	Total Hotels Identified from App	Total Images of Hotel Rooms in App	Total Global Downloads of App
Source			
1	145,000	1,500,000	

Traffick Cam 2021 vs. Global Hotel Numbers 2021

Table 27 Accuracy: Traffick Cam Metrics vs Global Metrics 2021

Total Global Numbers Compared to Traffick Cam App
2021

	Total Hotels	Total Global Downloads
Global Numbers	700,000	203 Billion
Traffick Cam Average	145,000	

RQ1b Implications: *Traffick Cam*'s data set reflects the data remains relatively constant so the null hypothesis is accepted and the alternate hypothesis is rejected. It is cautioned that future research in the area may include wider datasets that could strengthen or weaken the hypotheses findings.

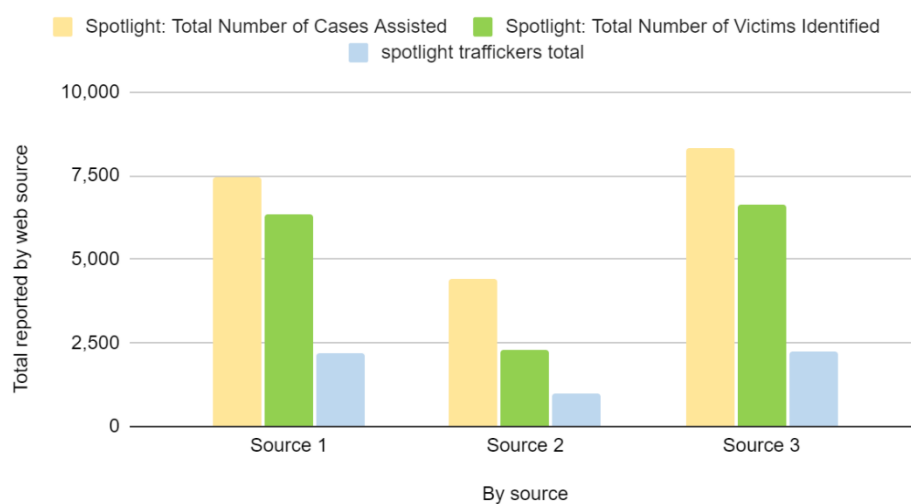
RQ2: Consistency

Consistency refers to how similar the metrics are in comparison in comparison to like AI platforms and then compared to national information. *Spotlight* was the only AI platform used because it included the most amount of information that could be compared to international (USA and Canada) information. The TIP reports from the Department of State and UNDOC were also compared to determine if they showed consistent data as well.

Spotlight 2016

Figure 16 Consistency: Spotlight Metrics 2016

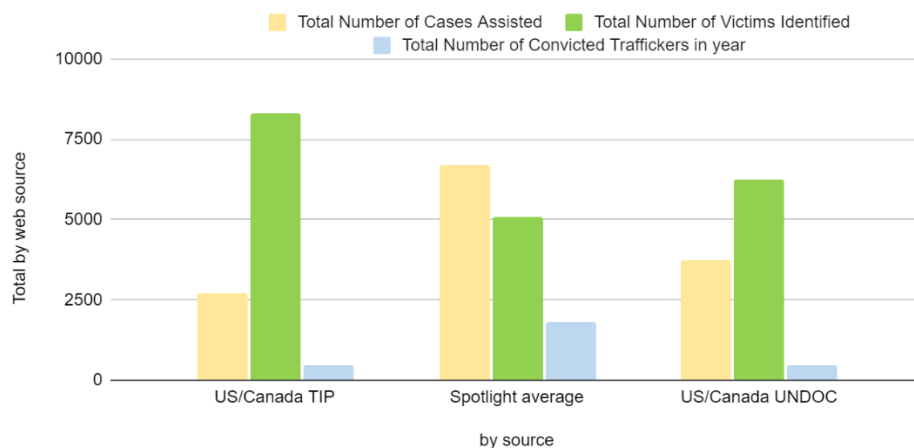
Spotlight: 2016



Consistency 2016: US/Canada TIP, US/Canada UNDOC, Spotlight

Figure 17 Consistency: Spotlight Metrics vs TIP Reports 2016

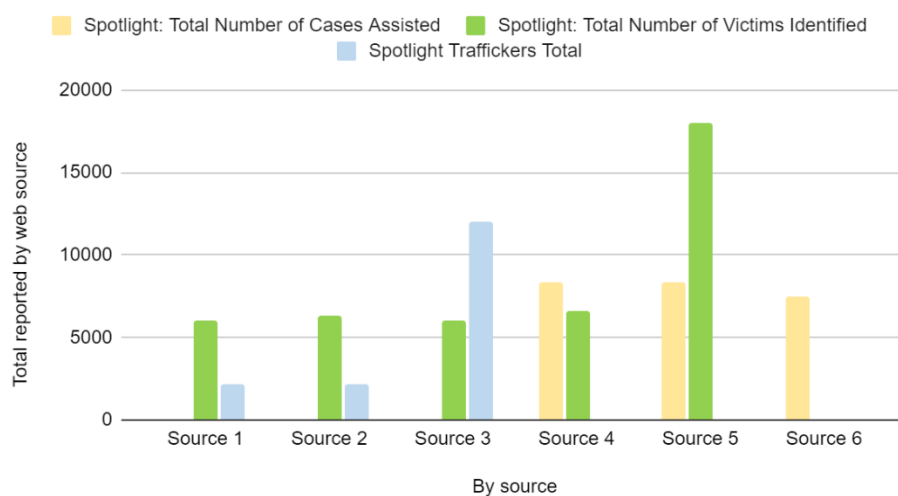
Consistency: 2016



Spotlight 2017

Figure 18 Consistency: Spotlight Metrics 2017

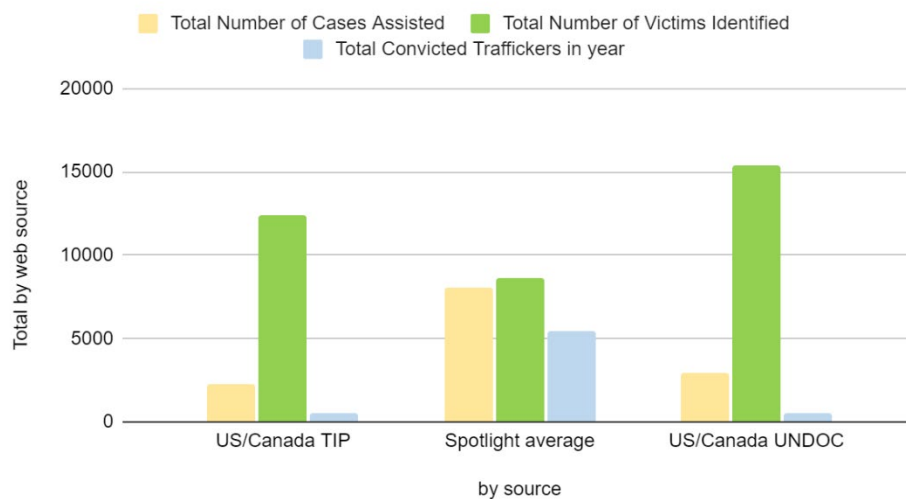
Spotlight: 2017



Consistency 2017: US/Canada TIP, US/Canada UNDOC, Spotlight

Figure 19 Consistency: Spotlight Metrics vs TIP Reports 2017

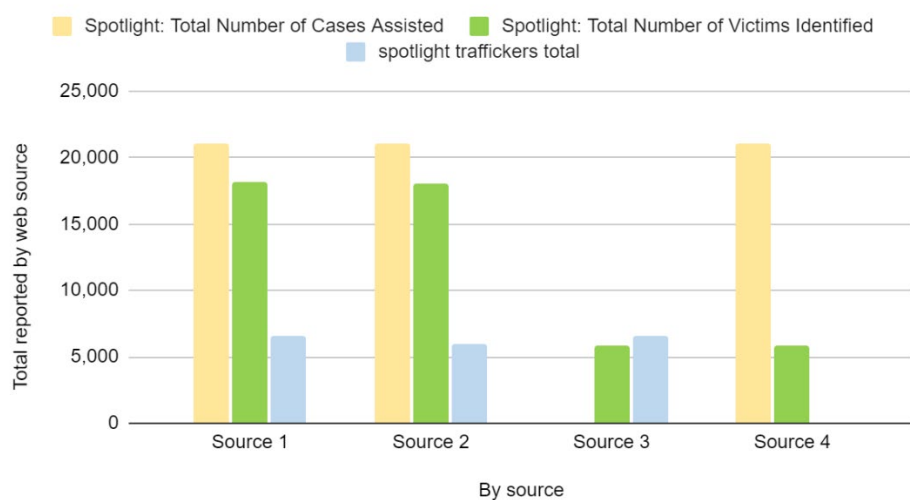
Consistency: 2017



Spotlight 2018

Figure 20 Consistency: Spotlight Metrics 2018

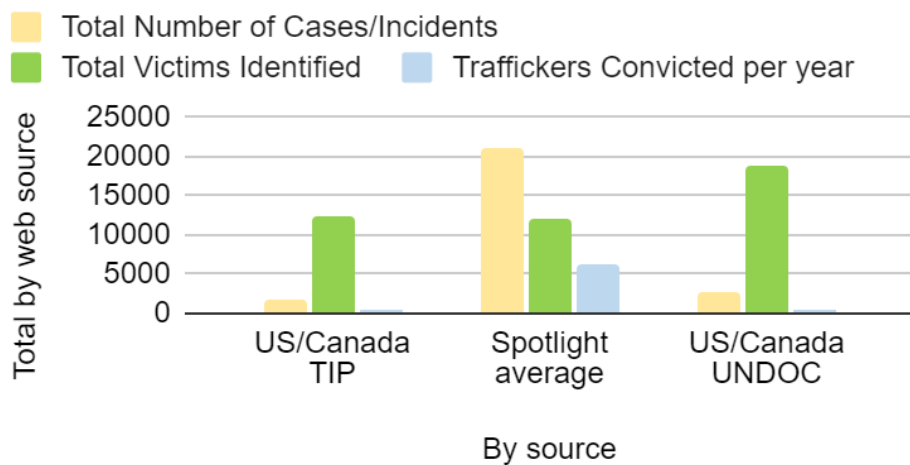
Spotlight: 2018



Consistency 2018: US/Canada TIP, US/Canada UNDOC, and Spotlight

Figure 21 Consistency: Spotlight Metrics vs TIP Reports 2018

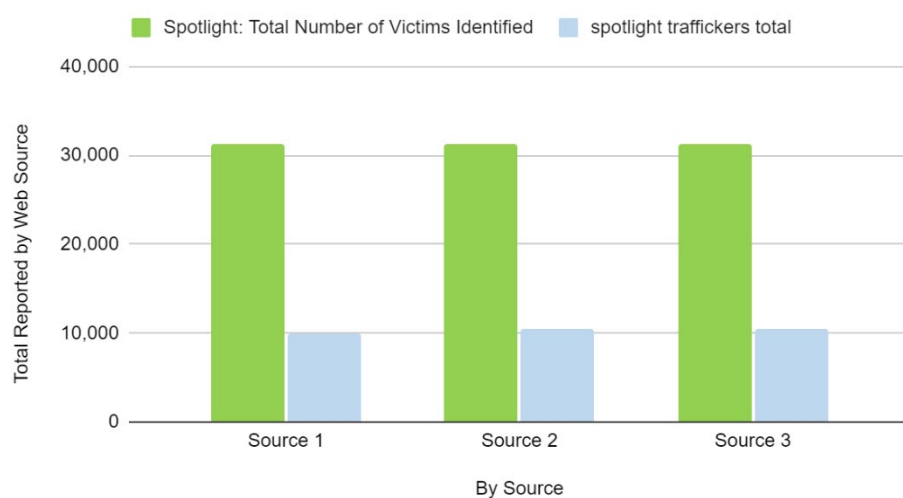
Consistency: 2018



Spotlight 2019

Figure 22 Consistency: Spotlight Metrics 2019

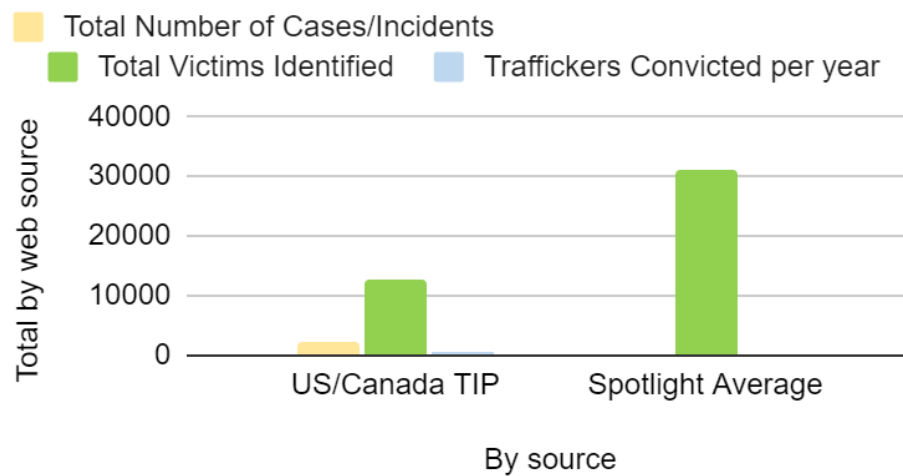
Spotlight: 2019



Consistency 2019: US/Canada TIP and Spotlight

Figure 23 Consistency: Spotlight Metrics vs TIP Reports 2019

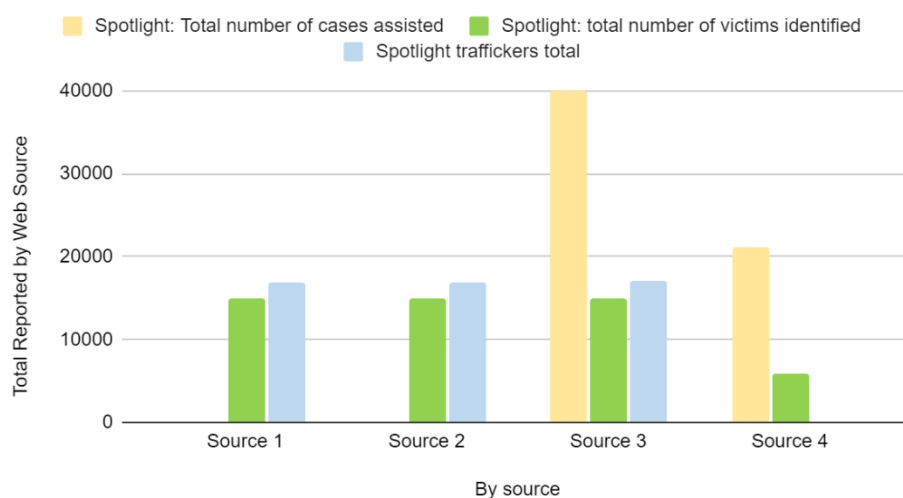
Consistency: 2019



Spotlight 2020

Figure 24 Consistency: Spotlight 2020

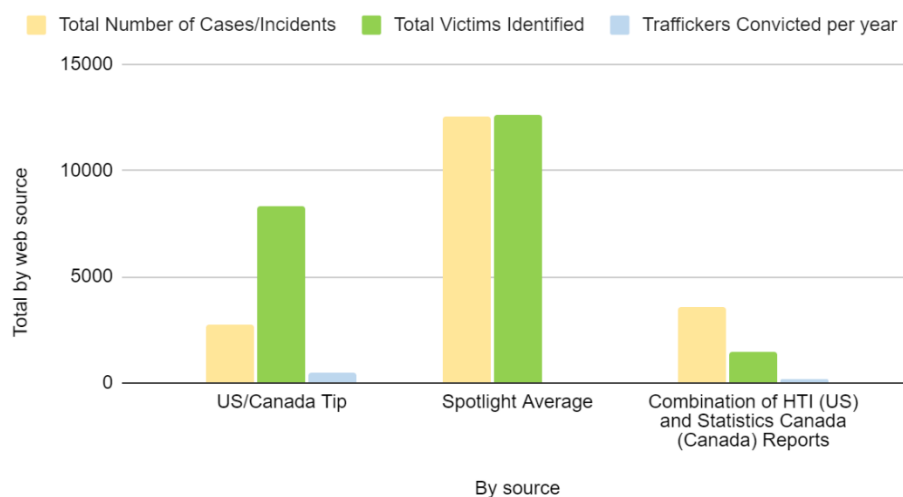
Spotlight: 2020



Consistency 2020: US/Canada TIP, Spotlight, and Combined HTI/Canadian Statistics

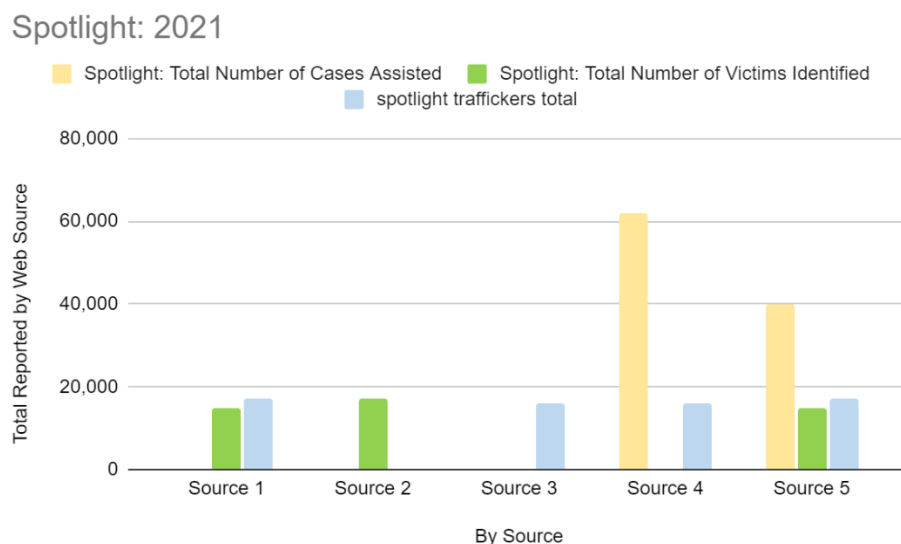
Figure 25 Consistency: Spotlight Metrics vs TIP Reports/Canadian Statistics Report 2020

Consistency: 2020



Spotlight 2021

Figure 26 Consistency: Spotlight Metrics 2021



Consistency 2021: Not enough data

RQ2 Implications: *Spotlight* was the only AI platform in this study because it had the most data and there was more available data that it could be compared to. As such, more complete datasets should be sought for future research to strengthen the hypotheses. Because *Spotlight* was the only AI platform used, the null and alternate hypotheses for *DIG*, *Traffick Jam*, and *Traffick Cam* are rejected. Inconsistencies in data and fluctuations throughout the years which created inconclusive results caused the null hypotheses and alternate hypothesis to be rejected regarding *Spotlight*. The researcher recommends future research in this area in the future to determine if these metrics change as more information is available.

RQ3: Effectiveness

The effectiveness is similar to the accuracy of the data but is directed towards how much the AI platforms have helped overall. While each AI platform has assisted in some way, it is important to determine which ones seem to assist more in combatting human trafficking so future

research can try to determine what the reasoning is that makes them more or less successful than other AI platforms.

RQ3: DIG

DIG 2016

No data to report

DIG 2016 vs. US 2016 TIP Report

Table 28 Effectiveness: DIG Metrics vs US TIP Report 2016

2016			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	2713	7956	439
DIG (Average)			

DIG 2017

Table 29 Effectiveness: DIG Metrics 2017

2017	DIG: Total Number of Cases Assisted	DIG : Total Number of Victims Identified	DIG: Total Convicted Traffickers
Source 1	3		
Source 2	3		

DIG 2017 vs. US TIP Report

Table 30 Effectiveness: DIG Metrics vs US TIP Report 2017

2017			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1,845	11,992	499
DIG (Average)	3		

DIG 2018

Table 31 Effectiveness: DIG Metrics 2018

2018	DIG: Total Number of Cases Assisted	DIG : Total Number of Victims Identified	DIG: Total Convicted Traffickers
Source 1	20		

DIG 2018 vs. US TIP Report

Table 32 Effectiveness: DIG Metrics vs US TIP Report 2018

2018			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1773	12552	526

DIG (Average)	20		
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DIG 2019

No data to report

DIG 2019 vs. US TIP Report

Table 33 Effectiveness: DIG Metrics vs US TIP Report 2019

2019			
	DIG: Total Number of Cases Assisted	DIG : Total Number of Victims Identified	DIG: Total Convicted Traffickers
TIP	1830	11976	475
DIG (Average)			

DIG 2020

No data to report

DIG 2020 vs. US TIP Report

Table 34 Effectiveness: DIG Metrics vs US TIP Report 2020

2020			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	1865	14571	309

DIG (Average)			
------------------	--	--	--

DIG 2021

No data to report

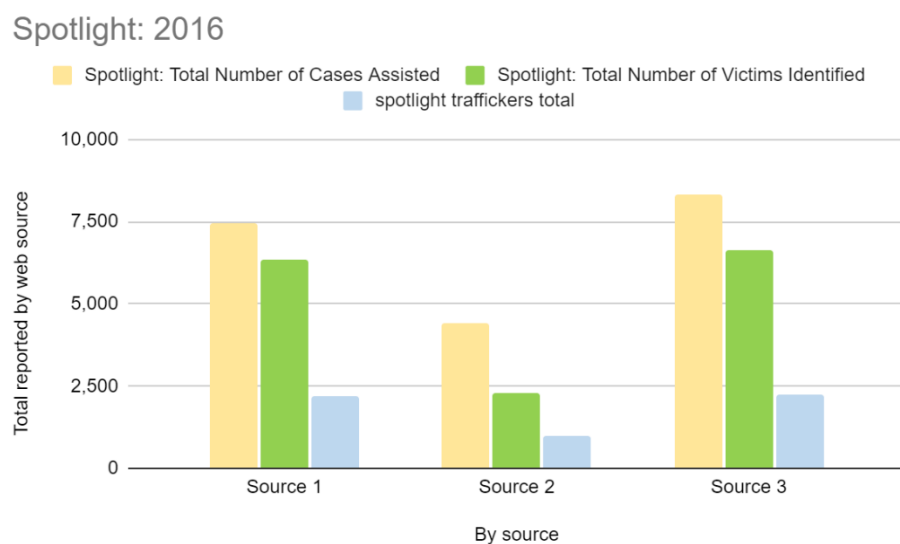
DIG 2021 vs. US TIP Report

No data to report

RQ3: Spotlight

Spotlight 2016

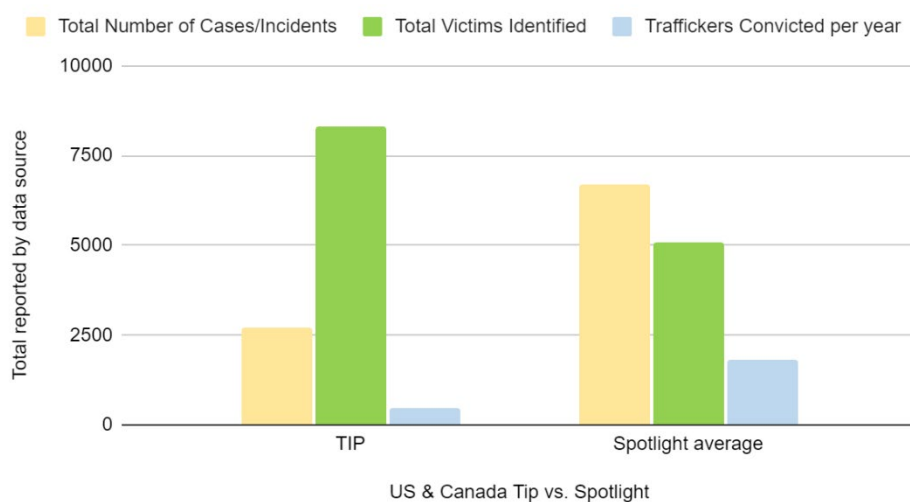
Figure 27 Effectiveness: Spotlight Metrics 2016



Spotlight 2016 vs. Combined 2016 TIP Reports (US and Canada)

Figure 28 Effectiveness: Spotlight Metrics vs combined US and Canada TIP Reports 2016

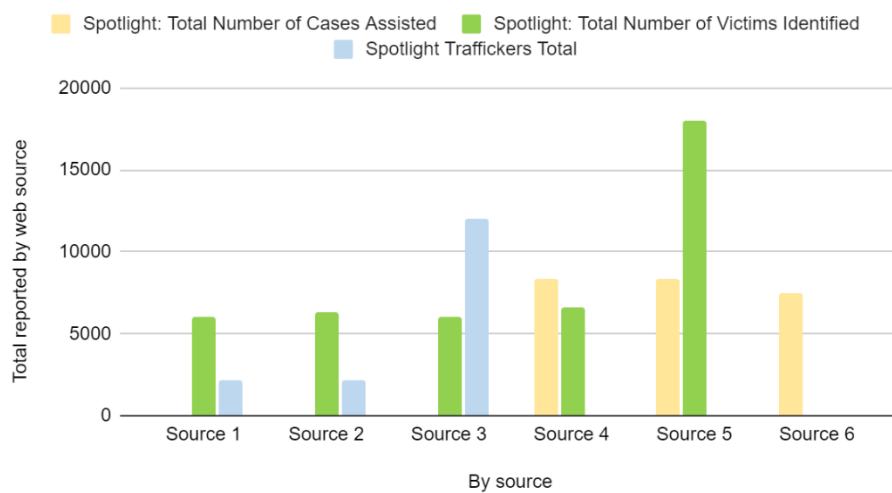
(2016) US & Canada TIP vs. Spotlight average



Spotlight 2017

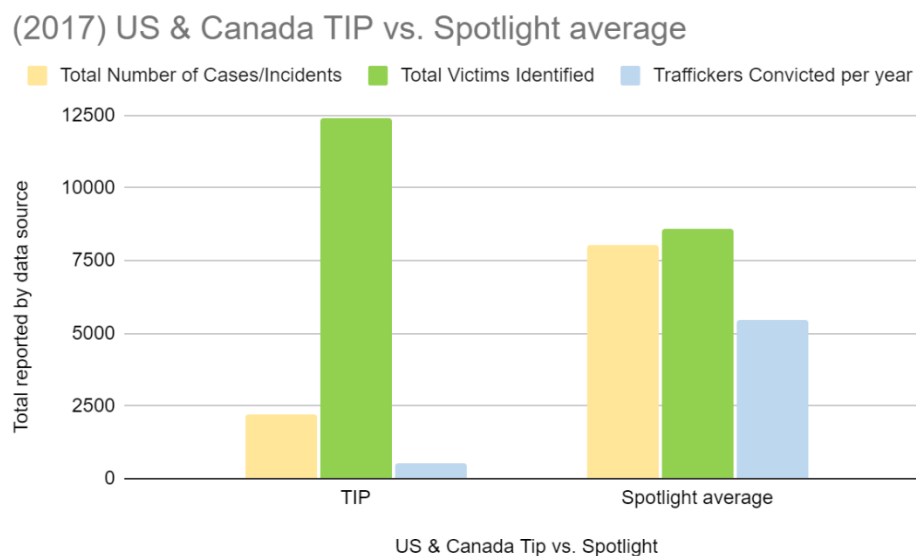
Figure 29 Effectiveness: Spotlight Metrics 2017

Spotlight: 2017



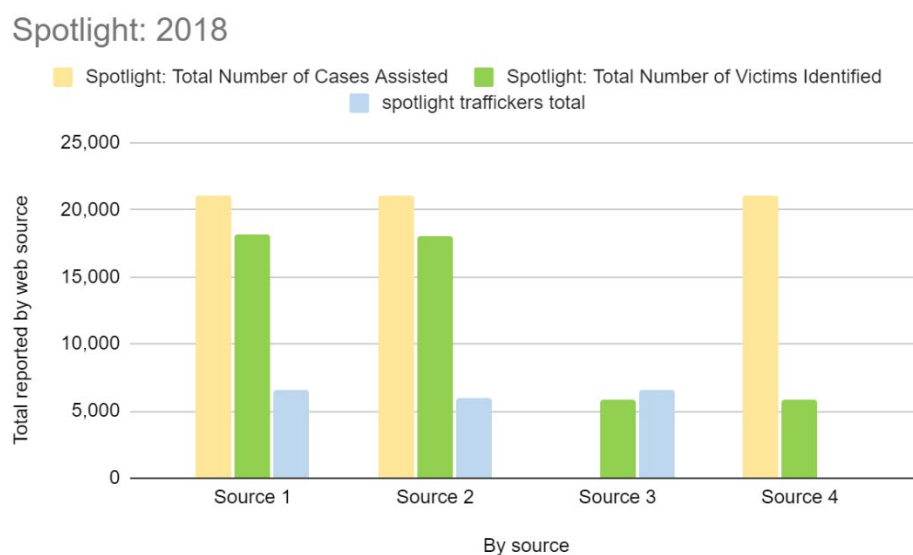
Spotlight 2017 vs. Combined 2017 TIP Reports (US and Canada)

Figure 30 Effectiveness: Spotlight Metrics vs combined US and Canada TIP Reports 2017



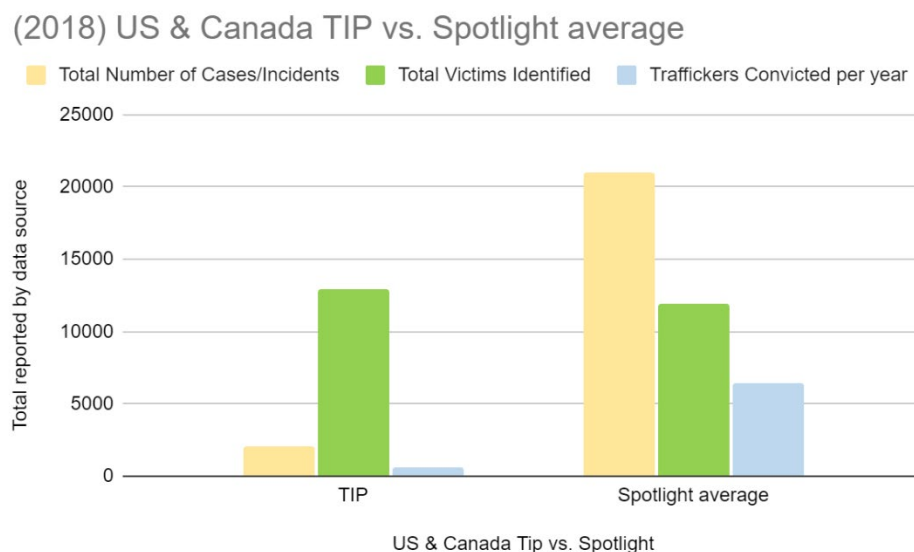
Spotlight 2018

Figure 31 Effectiveness: Spotlight Metrics 2018



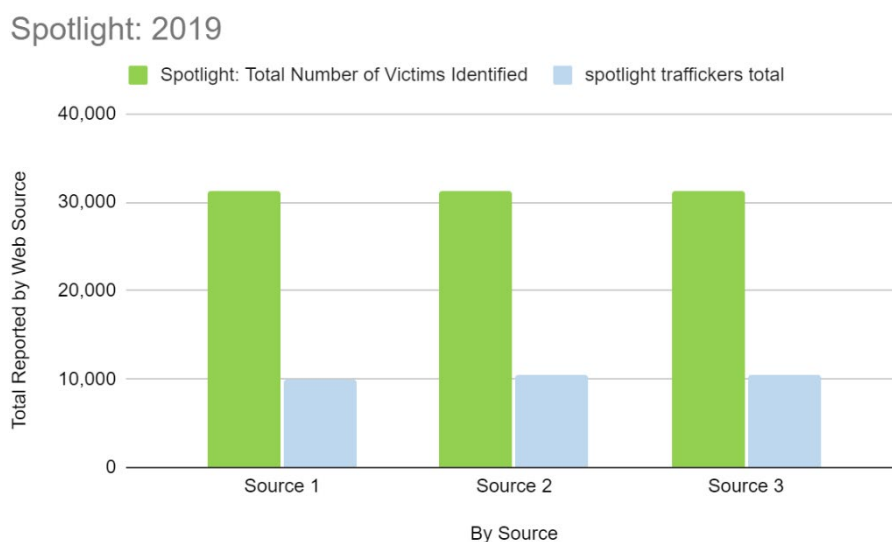
Spotlight 2018 vs. Combined 2018 TIP Reports (US and Canada)

Figure 32 Effectiveness: Spotlight Metrics vs combined US and Canada TIP Reports 2018



Spotlight 2019

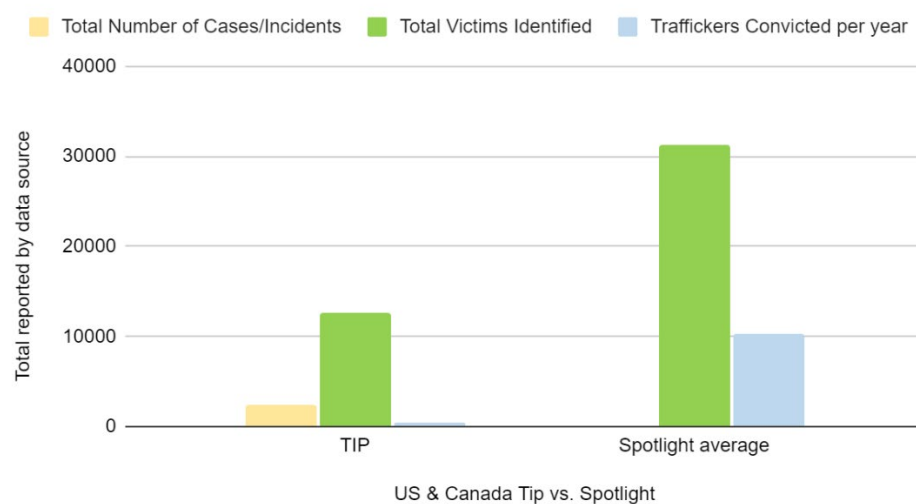
Figure 33 Effectiveness: Spotlight Metrics 2019



Spotlight 2019 vs. Combined 2019 TIP Reports (US and Canada)

Figure 34 Effectiveness: Spotlight Metrics vs combined US and Canada TIP Reports 2019

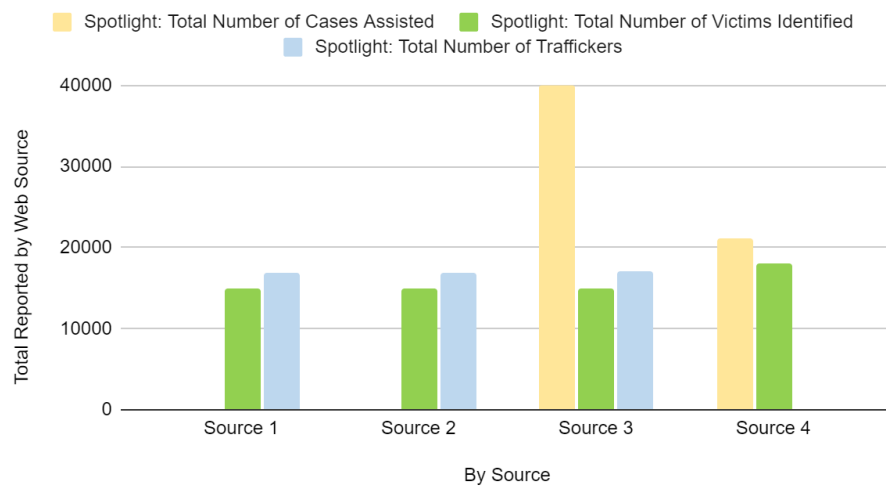
(2019) US & Canada TIP vs. Spotlight average



Spotlight 2020

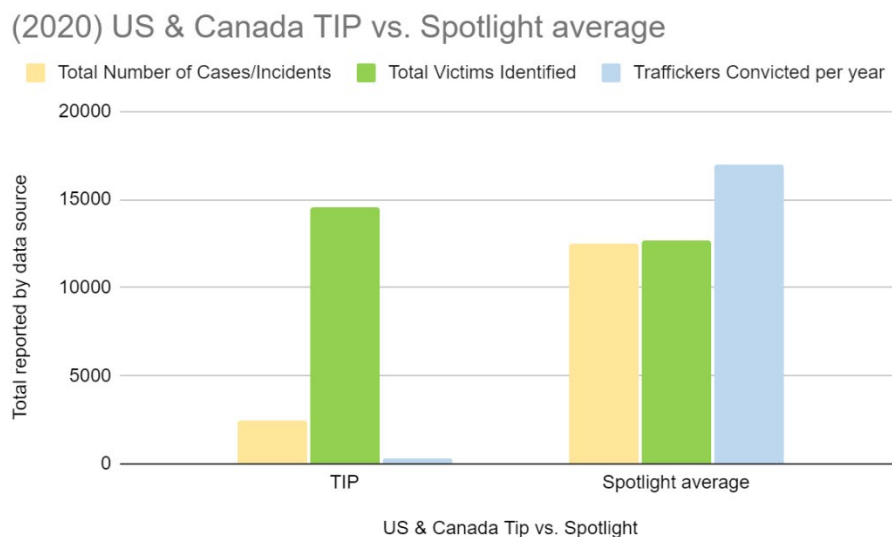
Figure 35 Effectiveness: Spotlight Metrics 2020

Spotlight: 2020



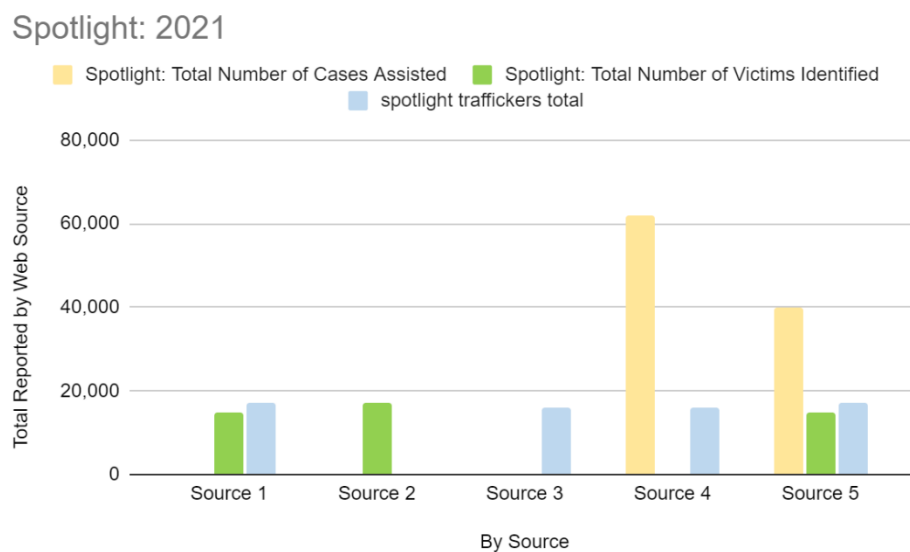
Spotlight 2020 vs. Combined 2020 TIP Reports (US and Canada)

Figure 36 Effectiveness: Spotlight vs combined US and Canada TIP Reports 2020



Spotlight 2021

Figure 37 Effectiveness: Spotlight Metrics 2021



Spotlight 2021 vs. Combined 2021 TIP Reports (US and Canada)

Table 35 Effectiveness: Spotlight Metrics vs combined US and Canada TIP Reports 2021

Combined US & Canada TIP vs. Spotlight
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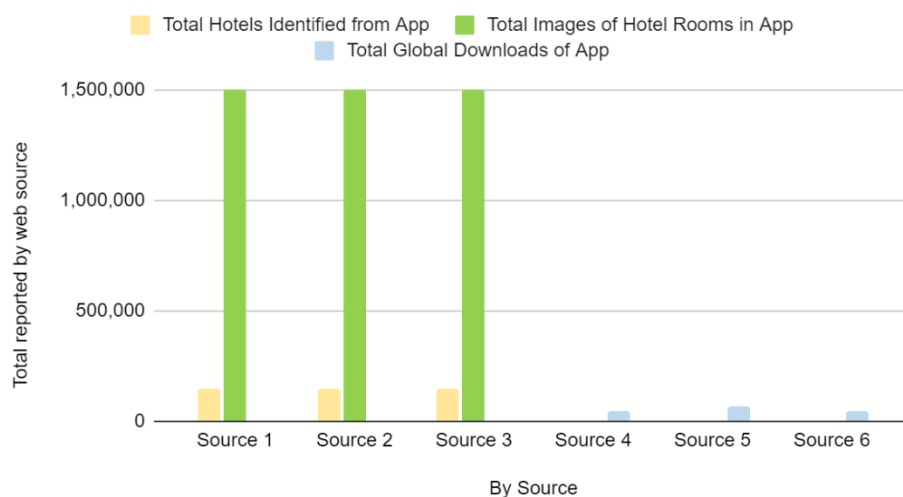
2021	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP	Not Available	Not Available	Not Available
Spotlight average	51,000	11,197	16,500

RQ3: Traffick Cam

Traffick Cam 2016

Figure 38 Effectiveness: Traffick Cam Metrics 2016

Traffick Cam (2016)



Traffick Cam 2016 vs. Global Hotel Numbers 2016

Table 36 Effectiveness: Traffick Cam Metrics vs Global Metrics 2016

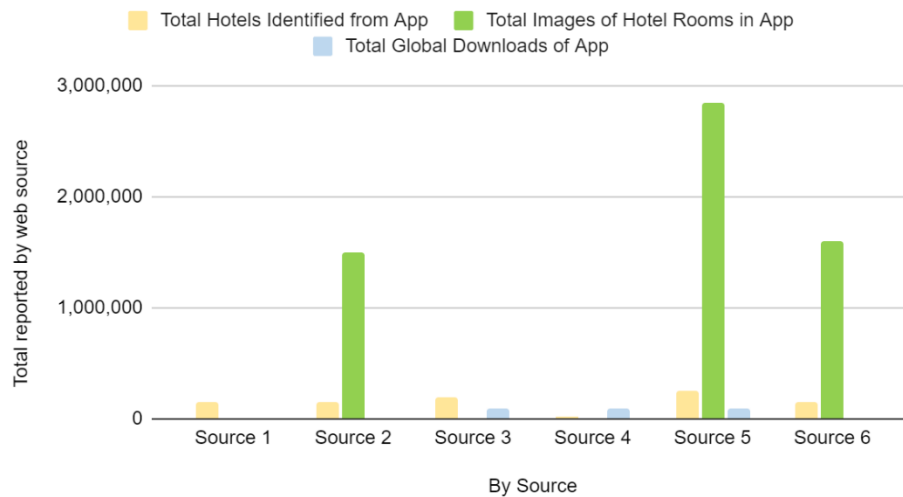
Total Global Numbers Compared to Traffick Cam App		
2016		
	Total Hotels	Total Global Downloads

Global Numbers	182,149	140.68 Billion
Traffick Cam Average	145,000	56,666

Traffick Cam 2017

Figure 39 Effectiveness: Traffick Cam Metrics 2017

Traffick Cam (2017)



Traffick Cam 2017 vs. Global Hotel Numbers 2017

Table 37 Effectiveness: Traffick Cam Metrics vs Global Metrics 2017

Total Global Numbers Compared to Traffick Cam App		
2017		
	Total Hotels	Total Global Downloads
Global Numbers		175 Billion
Traffick Cam Average	174,833	97,615

Traffick Cam 2018

Figure 40 Effectiveness: Traffick Cam Metrics 2018

Traffick Cam (2018)



Traffick Cam 2018 vs. Global Hotel Numbers 2018

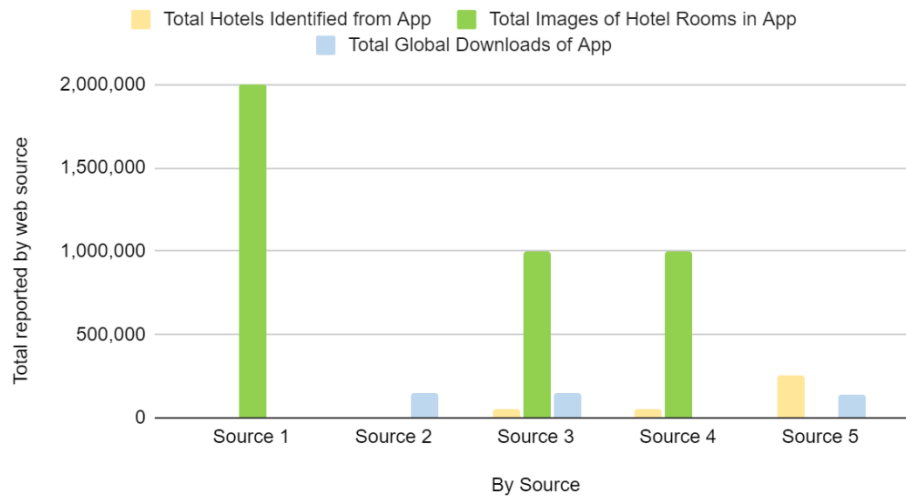
Table 38 Effectiveness: Traffick Cam Metrics vs Global Metrics 2018

Total Global Numbers Compared to Traffick Cam App		
2018		
	Total Hotels	Total Global Downloads
Global Numbers	184,229	192.45 Billion
Traffick Cam Average	97,500	100,000

Traffick Cam 2019

Figure 41 Effectiveness: Traffick Cam Metric 2019

Traffick Cam (2019)



Traffick Cam 2019 vs. Global Hotel Numbers 2019

Table 39 Effectiveness: Traffick Cam Metrics vs Global Metrics 2019

Total Global Numbers Compared to Traffick Cam App		
2019		
	Total Hotels	Total Global Downloads
Global Numbers	700,000	203 Billion
Traffick Cam Average	118,333	148,000

Traffick Cam 2020

Table 40 Effectiveness: Traffick Cam Metrics 2020

	Total Hotels Identified	Total Images of Hotel Rooms	Total Global Downloads
2020	from App	in App	of App

Source			
1	500,000	3,000,000	

Traffick Cam 2020 vs. Global Hotel Numbers 2020

Table 41 Effectiveness: Traffick Cam Metrics vs Global Metrics 2020

Total Global Numbers Compared to Traffick Cam App		
2020		
	Total Hotels	Total Global Downloads
Global Numbers	700,000	218 Billion
Traffick Cam Average	500,000	

Traffick Cam 2021

Table 42 Effectiveness: Traffick Cam Metrics 2021

2021	Total Hotels Identified from App	Total Images of Hotel Rooms in App	Total Global Downloads of App
Source			
1	145,000	1,500,000	

Traffick Cam 2021 vs. Global Hotel Numbers 2021

Table 43 Effectiveness: Traffick Cam Metrics vs Global Metrics 2021

Total Global Numbers Compared to Traffick Cam App
2021

	Total Hotels	Total Global Downloads
Global Numbers	700,000	203 Billion
Traffick Cam Average	145,000	

RQ3: Traffick Jam

Traffick Jam 2016

Table 44 Effectiveness: Traffick Jam Metrics 2016

2016	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Total Convicted Traffickers
Source 1		300	

Traffick Jam 2016 vs. US, Canada, and UK 2016 TIP

Table 45 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2016

2016			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	2,713	12,128	452
Traffick Jam		300	

Traffick Jam 2017

No data to report

Traffick Jam 2017 vs. US, Canada, and UK 2017 TIP

Table 46 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2017

2017			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	2,220	17,550	508
Traffick Jam			

Traffick Jam 2018

Table 47 Effectiveness: Traffick Jam Metrics 2018

2018	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Total Convicted Traffickers
Source 1		3,000	

Traffick Jam 2018 vs. US, Canada, and UK 2018 TIP

Table 48 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2018

2018			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year

TIP (US, UK & Canada)	3,237	19,965	582
Traffick Jam		3,000	

Traffick Jam 2019

Table 49 Effectiveness: Traffick Jam Metrics 2019

2019	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Total Convicted Traffickers
Source 1	600		
Source 2		3,800	

Traffick Jam 2019 vs. US, Canada, and UK 2019 TIP

Table 50 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2019

2019			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	3,820	23,179	531
Traffick Jam	600	3,800	

Traffick Jam 2020

Table 51 Effectiveness: Traffick Jam Metrics 2020

2020	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Total Convicted Traffickers
Source 1		3,800	
Source 2		6,800	
Source 3		6,800	

Traffick Jam 2020 vs. US, Canada, and UK 2020 TIP

Table 52 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2020

2020			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	4,283	25,184	316
Traffick Jam		5,800	

Traffick Jam 2021

Table 53 Effectiveness: Traffick Jam Metrics 2021

2021	Traffick Jam: Total Number of Cases Assisted	Traffick Jam : Total Number of Victims Identified	Traffick Jam: Total Convicted Traffickers
Source 1		3,800	
Source 2		3,800	
Source 3		3,800	

Traffick Jam 2021 vs. US, Canada, and UK 2021 TIP

Table 54 Effectiveness: Traffick Jam Metrics vs combined US, Canada, and UK TIP Reports 2021

2021			
	Total Number of Cases/Incidents	Total Victims Identified	Traffickers Convicted per Year
TIP (US, UK & Canada)	Not available	Not available	Not available
Traffick Jam		3,800	

RQ3 Implications: Again, *DIG* did not have many metrics and the numbers remained low which supports the null hypothesis and rejects the alternate. *Traffick Jam*’s data suggested the acceptance of the null hypothesis and the rejection of the alternate hypothesis. *Traffick Cam* was found to be inconclusive which rejected both hypotheses yet *Spotlight* seemed to improve

and the null hypothesis is rejected and the alternate hypothesis is accepted. Future research with more complete datasets could continue to support or challenge these findings.

Results

RQ1: Accuracy

RQ1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

H₀1: The accuracy of the following algorithms remained constant in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

RQ1a: *DIG, Spotlight, and Traffick Jam Specific:* According to the data documented in the descriptive statistics section, *DIG* held extremely low numbers for the years that data was available and supports the null hypothesis. The null hypothesis was also accepted regarding *Traffick Jam* because of the consistency in the data metrics. In the *Spotlight* data set regarding accuracy, *Spotlight* does not remain constant and improves and, therefore, the null hypothesis is rejected.

H_a1: The accuracy of the following algorithms has improved in a span of six years: *DIG, Spotlight, Traffick Jam, and Traffick Cam.*

RQ1a: *DIG, Spotlight, and Traffick Jam specific:* According to the data from *DIG*, there is a lack of information for each year and the alternate hypothesis is rejected. The alternative hypothesis is also rejected for the AI platform, *Traffick Jam*, because there is a lack of increase in the data points. *Spotlight* has improved according to the data set and, therefore, the alternate hypothesis is accepted.

H_01 : The accuracy of the following algorithms remained constant in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

RQ1b: *Traffick Cam* Specific: *Traffick Cam*'s data sets, on average, reflect that the data remains relatively constant in its execution which also indicates that the null hypothesis is accepted in this data set.

H_{a1} : The accuracy of the following algorithms has improved in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

RQ1b: *Traffick Cam* Specific: The alternate hypothesis is rejected regarding *Traffick Cam* due to the data remaining constant among the data set.

RQ2: Consistency

RQ2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

H_01 : The consistency rate of the following human trafficking AI platforms has remained constant in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

Spotlight was the only AI platform that could be used while studying consistency because it had the most data and there was more available data that it could be compared to. Therefore, the null hypothesis for *DIG*, *Traffick Jam*, and *Traffick Cam* are rejected. The researcher used data from the Trafficking in Person's (TIP) reports for the United States and Canada, the Federal Human Trafficking Institute's report (United States only), official Canadian statistics, and the TIP report that is available from the United Nations to compare the data with *Spotlight*'s findings of total victims assisted, total convicted traffickers, and total cases/incidents identified. The data set regarding *Spotlight* and these other areas of research support the decision that the null

hypothesis is rejected because of the inconsistencies between the data that has been reported in the databases and how *Spotlight* fluctuates through the years.

H_{a1}: The consistency rate for following human trafficking AI platforms has improved in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

Again, the alternate hypothesis could not be accepted for *DIG*, *Traffick Jam*, and *Traffick Cam* due to lack of information and areas to compare these AI platforms with. Using the data from the TIP reports for the United States and Canada, the Federal Human Trafficking Institute's report (United States only), official Canadian statistics, and the TIP report that is available from the United Nations to compare the data with *Spotlight*'s findings of total victims assisted, total convicted traffickers, and total cases/incidents identified, the alternative hypothesis is rejected due to many inconclusive results. While some years seem to be improving, there were outliers in the data which does not support the alternate hypothesis and therefore it was failed to be accepted.

RQ3: Effectiveness

RQ3: *In a span of six years, how effective has detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

H₀₁: The effectiveness of the following human trafficking AI platforms has remained constant in a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

RQ3: According to the data documented in the descriptive statistics section, *DIG* held extremely low numbers for the years that data was available and supports the null hypothesis, therefore, the null hypothesis is accepted. The null hypothesis is also accepted regarding the AI

platforms, *Traffick Jam* and *Traffick Cam*. In the *Spotlight* data set regarding accuracy, *Spotlight* does not remain constant and improves and, therefore, the null hypothesis is rejected.

H_{a1}: The effectiveness of the following human trafficking AI platforms have improved over a span of six years: *DIG*, *Spotlight*, *Traffick Jam*, and *Traffick Cam*.

RQ3: According to the data from *DIG*, there is a lack of information for each year with little improvement therefore the alternate hypothesis is rejected. Regarding *Traffick Jam* and *Traffick Cam*, again, the alternative hypotheses are rejected. *Spotlight* has improved according to the data set and, therefore, the alternate hypothesis is accepted.

Summary

This chapter reflects the data that was obtained for this dissertation. Depending on the question and data, the null or alternative hypotheses were accepted or rejected. Regarding the platforms that were researched, *Spotlight* and *Traffick Cam* had the most amount of data and could be studied more effectively. This may indicate that these are two platforms that are more accurate and effective than the others. While some information may be skewed because of the lack of information or outliers in the metrics that were found, it should be noted that *Spotlight* and *Traffick Cam* are great assets to law enforcement and human trafficking victims based on the data. In the next chapter, the results will be discussed with more information and deeper conclusions will be drawn from this data.

CHAPTER FIVE: CONCLUSIONS

Overview

This quantitative-comparative study examined the factors of accuracy, consistency, and effectiveness of four specific AI platforms to evaluate whether the factors increased, decreased, or remained static among the focal AI platforms across a six-year period (2016 - 2021). The AI platforms in focus were *Spotlight*, *DIG*, *Traffick Jam*, and *Traffick Cam*. The null hypothesis for each research question stated that the AI platform would remain the same while the alternate hypothesis for each research question stated that the AI platform would improve in the same six-year timeframe. This chapter of the dissertation presents the discussion based on the results of the study, as well as its implications, limitations, and the suggestions for future research.

Discussion

The AI platforms that were utilized in this study were *Spotlight*, *DIG*, *Traffick Jam*, and *Traffick Cam*. On the one hand, the research focused on rates specific to how many human trafficking cases/incidents were assisted per platform, how many victims were assisted per platform, and how many traffickers were convicted per platform regarding *Spotlight*, *DIG*, and *Traffick Jam*. On the other hand, *Traffick Cam* focused on rates of how many hotels were uploaded into the application, how many images were in the application, and how many global downloads there were of the application. Although the researcher attempted to collect three data points per areas in question for each AI platform, this could not be achieved with each AI platform and year because there is limited information at this time regarding AI's effects on human trafficking. Perhaps in the future, there will be more information to use in studies like this for a better understanding of how much AI assists in human trafficking cases/incidents.

In the next section, the researcher will briefly present an overview of the main findings, and this overview will be followed by interpretation of the findings. To explain the findings, the

researcher first lists the AI platforms' main findings, how each research question was answered regarding the hypotheses, and how the theories that were mentioned in the literature review could help explain the results.

Discussion: Brief Overview of Major Findings

DIG, which is the Domain Insight Graph made by the organization *Memex*, had the least amount of data to use because it was only used as a prototype in the span of less than a year. The data also reflected more information pertaining to categorizing advertisements in clusters which was not part of the research scope (Szekely & Kejriwal, 2018). Additionally, there were only three data points which indicates that *DIG*, as an AI platform, and the findings were not very successful in any of the research questions.

Following closely with a lack of data is *Traffick Jam*, the null hypotheses for Research Questions one and three: Accuracy and Effectiveness, was accepted because of a lack of data changes through the years. The alternative hypotheses for the same two questions was rejected. However, Research Question two focused on Consistency and because there was a lack of information, the researcher did not compare *Traffick Jam* with the TIP reports or other statistics and, therefore, both hypotheses are failed to be accepted.

Spotlight was the other AI platform that the researcher used to compare similar rates of number of cases/incidents a platform assisted, how many victims were assisted, and how many traffickers were convicted in this study. The researcher found the most data for *Spotlight* but some of the results could be skewed in the data. For example, McMaster (2017) quoted Ashton Kutcher stating "We've identified 6,000 children that are being abused...we've identified 12,000 adults that are also being abused" which is not what each source clarified. While *Spotlight* had more data sources, the founders have diverted their efforts more to a comprehensive type of

solution called *Safer* which aims to assist in the identification and removal processes associated with child sexual abuse material (CSAM) (Thorn, 2022).

The last AI platform that was researched in this study was *Traffick Cam* which compared rates of the total number of hotels identified in the application, the total amount of images of hotel rooms that have been uploaded into the application, and the total number of downloads of the application. All of these rates were based on information on a global scale and because the rates that were found on the internet were not similar to the ones of the other AI platforms, *Traffick Cam* could not be compared to the other AI platforms but could be compared to global numbers of the rates mentioned regarding this application. What the researcher found similar to other applications is that *Traffick Cam*'s developers has begun to shift direction around 2018-2019 and investigate newer technology called *Hotels50K*.

Discussion: Research Questions

The research questions pertained to the accuracy, consistency, and effectiveness of the AI platforms in this study through a quantitative-comparative analysis. All of the information was obtained through public online websites so the study can be repeated in the future. The research questions were answered with information in the following ways:

RQ 1: *To what extent has the accuracy rate of DIG, Spotlight, Traffick Jam, and Traffick Cam changed over a span of six years within their programs and in comparison to each other?*

Research Question 1a focused on the accuracy of *DIG*, *Spotlight*, and *Traffick Jam* while Research Question 1b focused on the accuracy of *Traffick Cam*. In the case of *DIG* and *Traffick Jam*, the numbers remained low and constant which supported the null hypothesis. *Spotlight* was the only AI platform that rejected the null hypothesis because it seemed to improve each year which accepted the alternate hypothesis of improvement. Research Question 1b was specific to

Traffick Cam's results in which the accepted the null hypothesis of a constant level and rejected the alternate hypothesis of improvement because the numbers remained fairly similar through the years of 2016-2021.

RQ 2: *Individually and in comparison to each other, how consistent are the rates of detection of human trafficking and/or apprehension in the following algorithms within a span of six years: DIG, Spotlight, Traffick Jam, and Traffick Cam?*

Research Question 2 focused on the consistency using *Spotlight* because it was the only AI platform with enough data points to compare with outside sources (TIP report, Human Trafficking Reports, UNDOC TIP report, and official Canadian statistics). *DIG*, *Traffick Jam*, and *Traffick Cam* could not be compared with these statistics because of a lack of data (*DIG* and *Traffick Jam*) that can be obtained and rates that could not be compared (*Traffick Cam*) therefore *Spotlight* was the only AI platform used in this research question. Through the comparisons, the researcher found the null hypothesis that focused on a constant level for *Spotlight* is rejected. The alternate hypothesis of improvement through the years must also be rejected because of the lack of data (*Traffick Jam* and *DIG*), incomparable rates (*Traffick Cam*), and severe outliers (*Spotlight*).

RQ3: *In a span of six years, how effective has detecting human trafficking victims and their traffickers through DIG, Spotlight, Traffick Jam, and Traffick Cam, been individually and in comparison to each other?*

Research Question 3 focused on the effectiveness of each AI platform through the use of all data points obtained for this study. This provided the same results as accuracy but can be explained further to fulfill the requirement of effectiveness. Effectiveness, in this sense, is defined as the specifics of the results and is used more for strategical purposes (Surbhi, 2018)

while accuracy is used more as a near-perfect agreement between the measured and true quantity of the values of this study (Vaidya, 2022). In this argument, accuracy is a broader overview while effectiveness is as true, to the average, as possible.

The null hypothesis for Research Question 3 was that the values remained constant and this was accepted while comparing the *DIG* and *Traffick Jam* platforms, which rejected the alternate hypotheses. *Spotlight* had the most data points to use in this study and seemed to improve each year by great margins and therefore, the null hypothesis was rejected and the alternate hypothesis was accepted. The data from *Traffick Cam* showed inconsistent results which failed to accept both hypotheses.

Theoretical Framework of Human Trafficking

Previously in this dissertation, the researcher mentioned theories that are thought to influence human trafficking, namely, the General Strain theory, Lifestyle Exposure Theory, and the Routines Activities Theory (See Chapter 2). Now, the researcher has compared their findings and made conclusions based on the results and compared them to the theories' basic ideas.

First, the General Strain theory stated that the behaviors of victims and forced criminality were major influences of the construction of the theory (Templeton, 2011). Because the researcher did not use specific demographics in this study, they cannot make a direct connection between how these victims were obtained, however, data was obtained from the reports for research question 2 which does dive farther into demographics of the victims if future researchers are interested in the data points and how they relate to the General Strain Theory. Additionally, the researcher cannot make a comparison with the Lifestyle Exposure Theory because it also focuses on the demographics of the victims and how their surroundings impact their susceptibility to be victimized while this study did not include any demographic

information (Lutya, 2017).

The researcher can make minimal assumptions to the Routines Activities Theory because it does not focus solely on the demographics of a person. Instead, it states that there must be a motivated offender, a lack of a capable guardian, and a target of interest or attraction to the offender to occur. In essence, because the crimes occurred and are continuing to occur, this study relates mostly with the Routine Activities Theory. As it does not focus on specific people who can be trafficked, this opens the platform that anyone can be trafficked at any time based on the criteria of the theory (Lutya, 2017).

Implications

These findings can assist in determining how AI can be used in the world now and for future implications regarding human trafficking incidents/cases. With rapidly growing technological advances, the time to act is now which means law agencies, governments, and everyone in the world must be aware of how crime is changing and what it means to them. The AI platforms are used to specifically assist in the identification of victims, locations, clusters, and other areas to assist law enforcement and victims in many ways.

In particular, these AI platforms can be used on the Clear Net and the Dark Web which makes them useful in combatting human trafficking because they can identify patterns quicker than a human being and save time and lives in the process. Additionally, developers are using them as base models for more, improved methods to assist victims and convict traffickers. Because the Dark Web is virtually untraceable, it is essential to use AI platforms to continue to assist in combating crimes such as human trafficking (Omar & Ibrahim, 2020 & Reid & Fox, 2020).

The desired result of this research is to provide additional information on the AI platforms that are available to law enforcement, continue development of better strategies, and provide additional training information to law enforcement agencies, youth programs, prosecutors, immigration agencies, and other entities that can provide the most assistance to victims and potential victims (NIJ, 2020). Because the Dark Net is virtually untraceable, human trafficking crimes have been impacted with invisible issues that these AI platforms can assist with as they help and protect victims and assist in the convictions of human traffickers (Reid & Fox, 2020).

Limitations

There will also be limitations to any study and human trafficking creates a web of limitations that are known, understood, and taken into consideration. According to the Canadian Center to end Human Trafficking (2021), most human trafficking cases result in dismissals, discharges, or withdrawn which does not assist victims, potential victims, or anyone in the world except human traffickers. The limits of this study include the scope, sampling issues, and the lack of comparative data of other studies.

The scope of this study may be too narrow because the information does not support every piece of information the researcher could use in a study. There was information that the researcher found important but did not fit into this analysis which is a note limitation.

The sampling size is also too narrow because the researcher could have widened the research to more sources per metric. In some tables, there are as many as six sources however this is because the researcher was more focused on specific metrics (i.e. how many cases were assisted, how many victims were assisted, and how many traffickers were convicted) instead of obtaining as many data points as possible. However, if the researcher aimed for as many data

points as possible, the sampling size could potentially be too wide but this did not seem to be the case while the researcher was conducting the study.

The last limitation pertains to the lack of similar studies. The results of this study were not compared with other studies because the researcher did not find any similar studies in the timeframe of research. In the future, more studies may be conducted and further analysis could indicate if this study holds true for future events.

Recommendations for Future Research

There are still questions in the field of human trafficking crimes and the crimes are often underreported, misreported, or unreported which adds to the issues of combatting the crime. While the researcher delved into the various AI platforms in their study, they found additional information that serves as future recommendations of the area. These areas include information regarding the lack of overall reporting, additional statistics, and the Polaris Project. To better understand the implications of these results, future studies could address these areas to determine what else can be accomplished to combat human trafficking with more accurate, consistent, and effective results.

There are many sources that combine human trafficking as a whole and may not reflect the different subsections of the crime. While human trafficking is often synonymous with sex trafficking, the other subsections can include labor trafficking of individuals and drug trafficking by individuals who are being trafficked themselves. Many crimes can be hidden under hotels, factories, marriage brokers, restaurants, and most surprisingly, adoption firms (ACOG, 2019).

The researcher recommends future research in the area of providing more information to the public such as broadcasting the presentation of Traffick Jam Live (TJL) and their newest campaign of “EYES 4 U” (Traffick Jam Live, 2021). “EYES 4 U” seems to be the next step of

public education and can educate the public of what signs to look for and specific hand signals that can assist when the victim will not, or cannot, speak for themselves (Traffick Jam, 2021 & Perdue, 2021).

The Polaris Project is a company that assists in identifying human trafficking victims in a variety of ways. One of the most successful ways that Polaris assists is through the Human Trafficking Hotline. According to the Polaris Project, human trafficking survivors have increased by almost 20% since the hotline was produced. In 2019, statistics revealed that the hotline received more text messages than phone calls which only increased the chances of rescuing victims of human trafficking. The hotline also identified 22,326 victims of human trafficking in the same year (Moskovska, 2021).

Unfortunately, numerous sources declared that children who are in foster homes or runaways were in an increased risk of human trafficking. The risk also increases for children who are abused, use illicit drugs or have alcohol dependencies, or who are members of the LGBTQ community (Moskovska, 2021 & Case Western Reserve University, 2022). In fact, approximately 74% of human trafficking victims revealed that they had been in the foster care system (Chains Interrupted, 2022). Collectively, the countries who have seen the most human trafficking in the world are Mexico, the Philippines, and the United States of America (Moskovska, 2021).

To date, each state in the United States of America reports human trafficking crimes. Approximately 75% of the victims are sold online through advertisements on the clear net and dark web at a rate of about 150,000 new advertisements each day. During the Covid-19 lockdown, these rates jumped substantially by approximately 22% while recruitment in other areas decline: adult clubs dropped by 46%, schools dropped by 38%, and foster homes decreased

by 70%. During this time, it was found that there was a 125% increase in human trafficking reports through Facebook and about 95% of an increase through the Instagram application (Polaris Project, 2022).

The lack of overall reporting in the area of human trafficking. Sources have global estimates that range anywhere from between 12.3million to 45.8million human trafficking victims (Barrick, K. & Pfeffer, 2021). While human trafficking is a TOC, the estimates are widely ranged because of the misreporting, underreporting, and overall lack of reporting. This is connected to the fear that victims feel when they are entrapped and how inadequate the training, if even provided, is for law enforcement, the private sector, schools, and other establishments. Training should include recognizing signs of victimization and understanding how to assist these victims of the crime(s) (ACOG, 2019).

It is unfortunate to declare that law enforcement officials still need more training and time to combat human trafficking with the rising numbers. With the rapidly-changing environment of the internet and other technological advances, the need for more time, funding, and training is pertinent to helping more victims. The Polaris Project (2022) found that less than 5% of law enforcement agencies have designated officers to investigate human trafficking cases. Furthermore, only about 18% of law enforcement officers have specified training requirements that pertain to human trafficking (Polaris Project, 2022). These metrics indicate that more training could assist in saving lives, convicting traffickers, and protect potential victims in the future.

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APPENDIX A: IRB Approval Letter**LIBERTY UNIVERSITY**
INSTITUTIONAL REVIEW BOARD

December 27, 2021

Jamie Graham
Jared Perry

Re: IRB Application - IRB-FY21-22-519 The Dark Web and Human Trafficking

Dear Jamie Graham and Jared Perry,

The Liberty University Institutional Review Board (IRB) has reviewed your application in accordance with the Office for Human Research Protections (OHRP) and Food and Drug Administration (FDA) regulations and finds your study does not classify as human subjects research. This means you may begin your project with the data safeguarding methods mentioned in your IRB application.

Decision: No Human Subjects Research

Explanation: Your study is not considered human subjects research for the following reason: It will not involve the collection of identifiable, private information from or about living individuals (45 CFR 46.102).

Please note that this decision only applies to your current application, and any modifications to your protocol must be reported to the Liberty University IRB for verification of continued non-human subjects research status. You may report these changes by completing a modification submission through your Cayuse IRB account.

If you have any questions about this determination or need assistance in determining whether possible modifications to your protocol would change your application's status, please email us at irb@liberty.edu.

Sincerely,

G. Michele Baker, MA, CIP
Administrative Chair of Institutional Research
Research Ethics Office

APPENDIX B: Instruments

DIG Data Collection

The researcher used the following expression for all searches regarding the data collection for DIG: “DIG” + “Memex” + “human trafficking.” The criteria was used in Google Scholar and the Google search engine. The researcher attempted to collection three data points for each year but could not find enough data points for each time frame.

DIG figures 2016 - 2020

Source Number DIG	Total Number of Cases Assisted	Total Number of Victims Identified	traffickers total
2016			
2017			
1	3		
2	3		
2018			
1	300		
2019			
2020			
2021			

Sources between 1/1/2016-12/31/2016

N/A

Sources between 1/1/2017-12/31/2017

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N/A

Sources between 1/1/2020-12/31/2020

N/A

Sources between 1/1/2021-12/31/2021

N/A

Spotlight Data Collection

The researcher used the following expression for all searches regarding the data collection for Spotlight: “Spotlight” + “Thorn” + “human trafficking.” The criteria was used in Google Scholar and the Google search engine. The researcher attempted to collection three data points for each year but could not find enough data points for each time frame.

Spotlight figures 2016 - 2020			
Year	Spotlight: Total Number of Cases Assisted	Spotlight: Total Number of Victims Identified	spotlight traffickers total
2016			
Source 1	7,442	6,325	2,186
Source 2	4400	2300	1000
Source 3	8305	6649	2249
2017			
Source 1		6,000	2,200
Source 2		6325	2186

Source 3		6000	12000
Source 4	8305	6649	
Source 5	8300	18000	
Source 6	7442		
2018			
Source 1	21,044	18,119	6,553
Source 2	21000	18000	6000
Source 3		5894	6608
Source 4	21044	5791	
2019			
Source 1		31,197	10,000
Source 2		31,197	10,496

Source 3		31,197	10,496
2020			
Source 1		14,874	16,927
Source 2		14,874	16,927
Source 3	40000	15,000	17,000
Source 4	21044	18,119	
2021			
Source 1		15,000	17,000
Source 2		17,092	
Source 3			16,000
Source 4	62,000		16,000
Source 5	40,000	15,000	17,000

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Source		
Number	Total Number of	Total Number of
Traffick	Cases Assisted	Victims Identified
Jam		
2016		
1		300
2017		
2018		
1		3000
2019		
1	600	
2		3,800
2020		
1		3,800
2		6,800
3		6800
2021		
1		3,800
2		3,800

3		3800
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Sources between 1/1/2016-12/31/2016

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Traffick Cam Data Collection

The researcher used the following expression for all searches regarding the data collection for Traffick Cam: “Traffick Cam” + “human trafficking.” The criteria was used in Google Scholar and the Google search engine. The researcher attempted to collection three data points for each year but could not find enough data points for each time frame.

<i>TRAFFICK CAM 2016 - 2021</i>			
Identifying...	total hotels Identified from App	total images of hotel rooms in App	Total Global Downloads of the App
2016			

Source 1	145,000	1,500,000	
Source 2	145,000	1,500,000	
Source 3	145,000	1,500,000	
Source 4			50,000
Source 5			70,000
Source 6			50,000
2017			
Source 1	150,000		
Source 2	145,000	1,500,000	
Source 3			92,846
Source 4	150,000		100,000
Source 5	254,000	2,850,000	100,000
Source 6	150,000	1,600,000	
2018			
Source 1	145,000	1,500,000	
Source 2	50,000	1,000,000	150,000
2019			
Source 1		2,000,000	
Source 2			150,000
Source 3	50,000	1,000,000	150,000
Source 4	50,000	1,000,000	
Source 5	255,000		144,000
2020			

Source 1	500,000	3,000,000	
2021			
Source 1	145,000	1,500,000	

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Data Reports for Comparison

US TIP Report

US TIP Report 2016-2021			
Year	Total Number of cases/incidents	Total number of open Victim cases that received assistance	Number of Convicted Traffickers per year
2016	2713	7956	439
2017	1845	11992	499
2018	1773	12552	526
2019	1830	11976	475
2020	1865	14571	309
2021			

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2020: U.S. Department of State (June 2021). Trafficking in Persons Report: United States. *United States Department of State*. 584-595. <https://www.state.gov/wp-content/uploads/2021/09/TIPR-GPA-upload-07222021.pdf>

US HTI Report

US HTI Report 2016-2021			
Year	Total Number of cases/incidents	Total number of open Victim cases that received assistance	Number of Convicted Traffickers per year

		(identified in cases)	
2016			
2017	783	1701	344
2018	771	1649	346
2019	606	1592	339
2020	579	1499	163
2021	607		238

2017: Feehs, K. E. & Richmond, J. C. (2018). 2017 Federal human trafficking report. *The Human Trafficking Institute*. <https://traffickinginstitute.org/wp-content/uploads/2022/01/2017-Federal-Human-Trafficking-Report-WEB-Low-Res.pdf>

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Human Trafficking Institute. [https://traffickinginstitute.org/wp-](https://traffickinginstitute.org/wp-content/uploads/2022/06/2021-Federal-Human-Trafficking-Report-Web.pdf)

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Canadian TIP Report

Canadian TIP Report 2016-2021			
Year	Total Number of Cases/Incidents	Total Victims Identified/Assisted	Traffickers Convicted per year
2016		367	10
2017	375	416	5
2018	340	420	51
2019	511	587	50
2020	573	(not reported)	(not reported)
2021			

2016: Department of State (2019). Trafficking in Persons Report June 2019. *United States*

Department of the State. [https://www.state.gov/wp-content/uploads/2019/06/2019-](https://www.state.gov/wp-content/uploads/2019/06/2019-Trafficking-in-Persons-Report.pdf)

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2019: U.S. Department of State (n.d.). 2021 Trafficking in Persons Report: Canada. *United States Department of State*. <https://www.state.gov/reports/2021-trafficking-in-persons-report/canada/>

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UK TIP Report

UK TIP 2016-2021			
Year	Total Number of cases/incidents	Total number of open Victim cases that received assistance	Number of Convictions/Convicted Traffickers per year
2016	(not reported)	3805	216
2017	(not reported)	5142	180
2018	1124	6993	191
2019	1479	10616	251
2020	1845	10613	197
2021			

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2020: U.S. Department of State. (2021). 2020 Trafficking in Persons Report: United Kingdom. *United States Department of State*. <https://www.state.gov/reports/2021-trafficking-in-persons-report/united-kingdom/>

Statistics Canada

Canadian Statistics 2016-2021			
Year	Total Number of cases/incidents	Total number of open Victim cases that received assistance	Number of Convicted Traffickers per year
2016			
2017			
2018			
2019			

2020	2977		
2021			

2020: Statistics Canada (2020). Trafficking in Persons in Canada, 2020. *Statistics Canada*.

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Global Report for Traffick Cam Comparison Data

GLOBAL REPORT 2016 - 2021			
GLOBAL REPORT 2016 - 2021	Hotels	Hotel Rooms	Total Number of Global App Downloads per year (in billions)
2016			
Source 1	182149		140.68
2017			
Source 1			175
2018			
Source 1	184299	16.97m	192.45
2019			
Source 1	700000	4m	204
2020			
Source 1	700000	16.4m	218
2021			
Source 1	700000	16.4m	203

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